

The Blockwise Rank Syndrome Learning problem and its applications to cryptography

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Abstract. This paper is an extended version of [8] published in PQCrypto 2024, in which we combine two approaches, blockwise errors and multi-syndromes, in a unique approach which leads to very efficient generalized RQC and LRPC schemes.

The notion of blockwise error in a context of rank based cryptography has been recently introduced in [31]. This notion of error, very close to the notion of sum-rank metric [27], permits, by decreasing the weight of the decoded error, to greatly improve parameters for the LRPC and RQC cryptographic schemes. A little before, the multi-syndromes approach introduced for LRPC and RQC schemes in [3, 18] also allowed to considerably decrease parameters sizes for LRPC and RQC schemes, through in particular the introduction of Augmented Gabidulin codes.

In order to combine these approaches, we introduced in [8] the Blockwise Rank Support Learning problem. It consists of guessing the support of the errors when several syndromes are given in input, with blockwise structured errors. The new schemes we introduced have very interesting features since for 128 bits security they permit to obtain generalized schemes for which the sum of public key and ciphertext is only 1.4 kB for the generalized RQC scheme and 1.7 kB for the generalized LRPC scheme.

In this extended version we give the following new features. First, we propose a new optimization on the main protocol which consists in considering 1 in the support of an error, allowing to deduce a subspace of the error to decode and improve the decoding capacity of our LRPC code, while maintaining an equal level of security. The approach of the original paper permits to reach a 40% gain in terms of parameters size when compared to previous results [18, 31], and this optimization allows to reduce the parameters by another 4% for higher security level. We obtain better results in terms of size than the KYBER scheme whose total sum is 1.5 kB. Second we give a more detailed analysis of the algebraic attacks on the ℓ -RD problem we proposed in [8], which allowed to cryptanalyze all blockwise LRPC parameters proposed in [31] (with an improvement of more than 40 bits in the case of structural attacks). And at last, third, we propose a more detailed introduction to the historical

background about rank metric, especially on the RQC and LRPC cryptosystems and their recent improvements and we add some parameters for the case of classical RQC (the case of only one given syndrome, that is a special case of our scheme, for which we could achieve 1.5 kB for the sum of the public key and the ciphertext), which compares very well to the previous version of classical RQC.

Keywords: code-based cryptography, Rank Syndrome Decoding problem, LRPC code, multiple syndromes, blockwise errors

1 Introduction and previous works

Background on rank metric code-based cryptography. Classical code-based cryptography relies on the Hamming distance but it is also possible to use another metric: the rank metric. This metric, introduced in 1985 by Gabidulin [19], is very different from the Hamming distance. In recent years, the rank metric has garnered significant attention from the coding community due to its relevance to network coding. Moreover, this metric can also be used for cryptography. Indeed, it is possible to construct rank-analogues of Reed-Solomon codes: the Gabidulin codes. These codes were used in early cryptosystems, like the GPT cryptosystem [20] which consists of an instantiation of the McEliece cryptosystem using Gabidulin codes. However, they were found to be inherently vulnerable due to the strong structure of the underlying codes. More recently, considering an approach similar to NTRU [24] (and also MDPC codes [26]), it became possible to construct a very efficient cryptosystem based on weakly structured rank codes: the LRPC cryptosystem [22]. Overall, the main interest of rank-metric based cryptography is that the complexity of the most known attack increases significantly with the size of the parameters: unlike Hamming code-based or lattice-based cryptography, it is possible to obtain a cryptosystem based on *a general instance of the rank decoding problem* with a size of only a few thousand bytes, while such parameter sizes can only be obtained with an additional structure (quasi-cyclic for example) in Hamming code-based or lattice-based cryptography. In the 2017 NIST standardization process, several schemes based on rank metric were proposed: LAKE, LOCKER, OUROBOROS-R and RQC. The three schemes LAKE, LOCKER and OUROBOROS-R were merged in the ROLLO 2nd round submission, while the RQC submission remained independent. Eventually, due to incertitude brought by algebraic attacks [14] that attacked NIST proposed parameters for rank metric, these schemes did not reach the 3rd round of the NIST standardization. However, the overall process permitted to reach a new audience for the potentiality of rank-based cryptosystems. The Loidreau cryptosystem [25] and its recent improvements [9] are further example of rank-based cryptosystem. In this paper, we focus on the LRPC and RQC cryptosystems.

Historical evolution of the LRPC cryptosystem. The main point that enables the LRPC cryptosystem to achieve small parameter sizes is its decoding algorithm. In the original 2014 version of the cryptosystem [22], the Decoding Failure Rate (DFR) is related to the block size n of the code, which is a major drawback when aiming for a very low DFR, as required to achieve IND-CCA2 security. The approach adopted for LRPC was either to consider a cryptosystem with a high DFR (around 2^{-30} , as in the LAKE cryptosystem), or considering a very low DFR, but at a cost of a high block size n , leading to very large parameters (as in the LOCKER cryptosystem). Overall, although the LAKE parameters were very appealing (public key $\simeq 600$ bytes), the high DFR remained a strong limitation. Conversely, achieving a very low DFR required such large parameters (4 kB) for LOCKER that the scheme became less competitive than its high-DFR counterpart. Another possibility for reducing the DFR was proposed in [10] but involves increasing m (the dimension of the extension field), which is generally too expensive. If one excepts the introduction of Ideal LRPC during the second round of NIST standardization process for ROLLO, which allowed to increase the number of choices for the block size of LRPC, there were no major breakthroughs for LRPC until the introduction in 2022 [3] of the multiple syndromes approach. This approach, based on the Rank Support Learning problem, permits to consider several syndromes. It has a strong impact on parameters since it permits to increase the number of considered syndromes and hence the overall decoding capacity of the code. This approach did not really change the high DFR approach, but had a major impact on the very low DFR approach which reached a size (pk+ct) of 2.4 kB, a strong improvement compared to the previous 4 kB. In practice, the multiple syndrome approach permits to consider a decoding capacity potentially close to the rank Gilbert-Varshamov bound which has a double impact on parameters: first, the complexity of attacks increases; second, approaching the RGV bound brings the scheme into a parameter space where algebraic attacks are less effective, with a complexity comparable to that of combinatorial attacks. The previously cited paper [3] also introduces unstructured LRPC variations of the scheme with very low parameters of 7 kB, beating the best unstructured lattices schemes. Finally, the paper also introduced the extended multiple syndromes (xMS) approach, which, at the cost of a slower decoding algorithm, allows LRPC codes to be decoded with smaller m , a crucial factor for achieving smaller parameters. Very recently, another approach was proposed in [31]. This approach uses blockwise errors to increase the decoding capacity of the LRPC codes: it allows to reach smaller parameters, but not as small as the multiple syndrome approach, primarily because the classical LRPC approach relies on large block size to reach very low DFR.

Historical evolution of the RQC cryptosystem. The RQC cryptosystem was submitted to the 2017 NIST standardization process and in [1], pre-published in 2016. It is also covered by the 2010 Gaborit-Aguilar patent [4]. The scheme is an equivalent in rank metric of the HQC scheme, which was also submitted to the NIST standardization process. The security of the protocol can be reduced to

the security of random instances, but it comes at a cost of two-parts ciphertexts, which naturally results in a larger parameter size. The main strong feature of the RQC protocol is its zero DFR thanks to the Gabidulin decoder, avoiding potential DFR existential drawbacks. In practice, the RQC parameters were rather large, and reached 5.6 kB (for 128 bits security) for public key and ciphertext size after algebraic attacks of 2019 [14]. There are two main reasons for this. First, the weight of the decoded error increases quadratically, which requires a larger block size n , and consequently a larger m (since m must be greater than n in Gabidulin codes). Second, the security of the RQC scheme is reduced to attacking a $[3n, n]$ code rather than a $[2n, n]$ code (as for LRPC), which significantly impacts the complexity of attacks. Overall, while the zero DFR is an attractive feature, the parameter size is less so. After the NIST submission, several improvements were proposed. First, in 2019, the concept of non-homogeneous error was introduced for the second-round submission of RQC. By sampling a common error support for the first $2n$ coordinates and a different support for the last n -length block, this approach aimed to address the costly $[3n, n]$ reduction. Finally, recently in [15], the notion of multiple syndromes was extended to the RQC cryptosystem. As with LRPC, this approach is very interesting in itself, but is even more efficient with the Augmented Gabidulin codes, also introduced in [15]. Augmented Gabidulin codes correspond to Gabidulin codes with additional zero positions, allowing in practice to mitigate the condition $n \leq m$. While this induces a non-zero DFR, the negative quadratic exponent makes the approach very efficient, as it allows for a reduction in m while maintaining similar decoding capacity and a very low DFR. This method, when combined with the multiple syndromes approach and non-homogeneous errors, enables parameter sizes to be reduced to 2.7 kB. It also permits to reach low parameters in the unstructured case (see [15] for details).

Recent results and introduction of blockwise rank errors for rank codes for LRPC and RQC schemes. Very recently in [31], the authors introduced the notion of rank blockwise errors, which allows for a reduction in the weight of decoded errors. The main idea of this approach is to consider words composed of blocks of respective length $n_1, \dots, n_\ell$ with each block associated with an error e_i of rank r_i with support E_i , such that the supports E_i intersect only at 0. In the case of $\ell = 2$, this allows for an error to be decoded in LRPC with a smaller weight $r_1 \cdot d_1 + r_2 \cdot d_2$, rather than $r \cdot d$ as in the classical LRPC case. In fact, to give a general idea, one exchanges the complexity of searching for an error of weight $2r$ and length $2n$ for the complexity of searching for a blockwise error of weight (r, r) associated with two blocks of length n . If one considers $r = d$ and $r_1 = r_2 = d_1 = d_2 = \frac{r}{2}$, the classical LRPC approach with homogeneous errors gives a syndrome of weight $r \cdot d = r^2$, whereas in the case of blockwise error the syndrome would have weight $r_1 \cdot d_1 + r_2 \cdot d_2 = \frac{r^2}{2}$. Decoding errors of smaller weight can have a significant impact on decoding performance. In their paper [31], the authors generalize previously known attacks to the blockwise rank error case (for both combinatorial and algebraic attacks), building on

recent results regarding non-homogeneous errors. They show that considering the blockwise approach rather than the classical homogeneous approach may be advantageous in some cases. This approach is especially interesting for the RQC scheme, for which they propose parameters with size 2.5 kB (public key + ciphertext), and somewhat less for the ILRPC case: with high DFR 2^{-30} , their parameters are 15% smaller than ROLLO-I (ex-LAKE, although we will later explain that their proposed parameters can be broken). Overall, the approach they propose is very interesting and fully develops the potential of rank metric.

Blockwise rank errors: why this new error structure is completely suited for rank metric based cryptography. As a well known notion, the rank metric benefits from strange properties. Indeed, suppose one aims to solve the RSD problem: $H.e^t = s$ (for e a codeword of $\mathbb{F}_{q^m}^n$ of weight r and H a random $(n - k) \times n$ matrix). In practice, the complexity of best attacks becomes linear whenever n becomes large enough. This property is directly related to the notion of support of the error: when the error length increases, the support of the error does not change. This peculiar property leads to the fact that it is easily possible to construct simple codes which can decode up to the rank Gilbert-Varshamov bound [21]. It is important to note that this feature is absent in Hamming or Euclidean distance. This property also explains why a straightforward adaptation of the Learning Parity with Noise (LPN) or Learning With Errors (LWE) problem does not work for rank metric: the system can be solved in polynomial time after a quadratic number of given syndromes. A way to obtain an equivalent approach for LPN or LWE in rank metric is proposed in [17]: instead of adding errors with the same support, one adds fixed-length block errors with varying error supports. This Learning with Rank Errors (LRE) approach permits to get an equivalent notion to LPN and LWE. The previous LRE approach is very close to the approach proposed in [31] and is also closely related to the sum-rank approach. The non-homogeneous approach of [15] can also be seen as a particular case of blockwise rank errors. In practice, the rank blockwise error approach permits to efficiently counter the attack in which, for a given m , one dramatically increases the length n of the code. The best combinatorial attacks have a complexity with roughly an exponent in krm/n . The blockwise structured error support counters the m/n effect, so that the best attacks essentially remains in kr for the exponent. This type of structured error is especially resistant for $[\ell n, n]$ codes with blocks of size n and $m = n$. This type of parameters is very well suited for ideal LRPC and RQC schemes, where the primary attacks directly correspond to this case. However, the case of unstructured schemes when m is larger than n (which is small) does not permit to benefit from the advantage of this blockwise structure, and no significant improvement seems to be achieved. Moreover, as explained in [31], the blockwise structure permits to decrease the weight of the error to decode in LRPC and RQC. This perspective suggests that the blockwise rank error approach is the natural one to adopt for rank metric: it naturally permits to get smaller error weights to decode and since, and is naturally resilient to the very long length attack approach which necessary

leads to polynomial attacks. This approach is especially efficient for RQC, as it counters the $[3n, n]$ attack that strongly impacts parameters. This explains why RQC parameters of [31] are rather small. In practice, this block size approach is especially interesting for the case where the main attack arises for $m \ll n$, which is precisely the case for ideal LRPC and RQC.

Contributions. In this paper, we combine the two previous approaches: multiple syndromes (along with Augmented Gabidulin codes) and blockwise errors for LRPC and RQC schemes. This new combined approach is especially efficient for the RQC scheme, allowing us to achieve parameters of 1.4 kB (public key + ciphertext) for 128-bit security, as the blockwise approach counters the $[3n, n]$ security reduction. However the approach in the case of LRPC codes combined with the xMS approach of [3] also remains interesting with a 1.7 kB size. These results represent a significant improvement, with a 40% reduction in parameter size compared to previous results, yielding parameters even smaller than KYBER (1.5 kB). It is the first time that one gets so small parameters in rank metric (and codes in general), along with very low DFR.

In addition to these main results, the contributions are as follows:

- We define a new problem: the Blockwise Rank Syndrome Learning problem, which enables the design of new generalized LRPC and RQC schemes using multiple syndromes and blockwise rank error approaches. We also generalize the xMS approach of [3] for the case of rank block errors.
- We propose new attacks for the ℓ -RD blockwise error problem, in specifically breaking all parameter sets of [31] for their LRPC variations. Notice that it does not alter the confidence we can have in the scheme, since parameters can be increased to counter this attack.
- We provide generalized combinatorial and algebraic attacks for the new Blockwise Rank Syndrome Learning problem.
- We revisit some combinatorial and algebraic attacks described in [31].

New contributions of this extended version:

- A new protocol optimization: we propose for our RQC-Block-MS-AG scheme to sample an error with 1 in the support of the block $\mathbf{R}_2$. It permits to reduce the Decoding Failure Rate of the Augmented Gabidulin code and thus reducing the parameters size further by almost 10%, without altering the practical security of the scheme.
- Algebraic attacks: we give further technical details on the analysis of the algebraic attacks on the ℓ -RD problem compared to the original version.
- Introduction: we give a more detailed introduction for the paper especially regarding the historical aspects and evolution of schemes in rank metric which permits to have a better general overview on this field.

Organisation of the paper. Section 1 gives a general overview of the situation for LRPC and RQC schemes and also gives a perspective on the blockwise rank error approach. Section 2 presents the general background on rank metric and cryptographic schemes. Section 3 describes the new blockwise RSL problem together with the generalization of the xMS approach in the case of blockwise rank errors. Section 4 gives a description of our new generalized RQC and LRPC schemes. Sections 5 and 6 present the details of combinatorial and algebraic attacks for the problem we address, while also revisiting some complexity aspects from [31]. Section 7 discusses the cryptanalysis of the LRPC parameters from [31]. Section 8 introduces new parameters based on our approach and compares them to other schemes.

2 Preliminaries

2.1 Background on the rank metric

Definition 1 (Rank metric over $\mathbb{F}_{q^m}^n$). For a vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_{q^m}^n$, we define the support $\text{Supp}(\mathbf{x}) \stackrel{\text{def}}{=} \langle x_1, \dots, x_n \rangle_{\mathbb{F}_q}$. The rank weight of $\mathbf{x}$ is equal to $\|\mathbf{x}\| \stackrel{\text{def}}{=} \dim(\text{Supp}(\mathbf{x}))$.

In the following, the set of vectors in $\mathbb{F}_{q^m}^n$ of rank weight r will be denoted by:

$$\mathcal{S}_r^n(\mathbb{F}_{q^m}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{F}_{q^m}^n \mid \|\mathbf{x}\| = r\}.$$

We will also use

$$\mathcal{S}_{r,1}^n(\mathbb{F}_{q^m}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{F}_{q^m}^n \mid \|\mathbf{x}\| = r, 1 \in \text{Supp}(\mathbf{x})\}.$$

Definition 2 ($\mathbb{F}_{q^m}$ -linear code). An $\mathbb{F}_{q^m}$ -linear code of parameters $[n, k]_{q^m}$ is an $\mathbb{F}_{q^m}$ -subspace of $\mathbb{F}_{q^m}^n$ of dimension k .

Such a code $\mathcal{C}$ can be represented by a full-rank generator matrix $\mathbf{G} \in \mathbb{F}_{q^m}^{k \times n}$ or by a full-rank parity-check matrix $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times n}$.

2.2 Rank Decoding and Rank Support Learning problems

The decoding problem relevant for all rank-based constructions is:

Definition 3 (RD Problem). Given $(\mathbf{G}, \mathbf{y}) \in \mathbb{F}_{q^m}^{k \times n} \times \mathbb{F}_{q^m}^n$, the Rank Decoding problem $\text{RD}(n, k, r)$ asks to compute $\mathbf{e} \in \mathbb{F}_{q^m}^n$ such that $\mathbf{y} = \mathbf{x}\mathbf{G} + \mathbf{e}$ and $\|\mathbf{e}\| \leq r$. We will write RSD for the equivalent version written with a parity-check matrix.

Even if RD is not known to be NP-complete, [29] gives a randomized reduction to the decoding problem in the Hamming metric, this time NP-complete. The Rank Support Learning problem [21] is a generalization of RD where we are given N instances with the same generator matrix (or the same parity-check matrix for RSD) and where the errors have the same support.

Definition 4 (RSL Problem). Given $(\mathbf{H}, \mathbf{S}) \in \mathbb{F}_{q^m}^{(n-k) \times n} \times \mathbb{F}_{q^m}^{N \times (n-k)}$, the Rank Support Learning Problem $\text{RSL}(n, k, r, N)$ asks to compute a subspace $E \subset \mathbb{F}_{q^m}$ of dimension r for which there exists a matrix $\mathbf{V} \in E^{\ell \times n}$ such that $\mathbf{H}\mathbf{V}^\top = \mathbf{S}^\top$.

2.3 Ideal codes

Let $P \in \mathbb{F}_q[X]$ be an irreducible polynomial of degree n . We define the internal product of two vectors $\mathbf{x}, \mathbf{y}$ in $\mathbb{F}_{q^m}^n$ as $\mathbf{x} \cdot \mathbf{y} \stackrel{\text{def}}{=} \mathbf{X}(X)\mathbf{Y}(X) \bmod P$, where $\mathbf{X}(X) = \sum_{i=0}^{k-1} x_i X^i$ and $\mathbf{Y}(X) = \sum_{i=0}^{k-1} y_i X^i$.

Definition 5 (Ideal matrix). Let $P \in \mathbb{F}_q[X]$ be a polynomial of degree n and let $\mathbf{v} \in \mathbb{F}_{q^m}^n$. The ideal matrix generated by $\mathbf{v}$ and P , denoted by $\mathcal{IM}_P(\mathbf{v})$ (or $\mathcal{IM}(\mathbf{v})$ if there is no ambiguity on P), is the element of $\mathbb{F}_{q^m}^{n \times n}$ defined by

$$\mathcal{IM}_P(\mathbf{v}) \stackrel{\text{def}}{=} \begin{pmatrix} \mathbf{v}(X) \bmod P \\ X\mathbf{v}(X) \bmod P \\ \vdots \\ X^{k-1}\mathbf{v}(X) \bmod P \end{pmatrix}.$$

One can see that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}\mathcal{IM}(\mathbf{v}) = \mathbf{v}\mathcal{IM}(\mathbf{u}) = \mathbf{v} \cdot \mathbf{u}$, so that the internal product is a matrix-vector product by the ideal matrix. An ideal code of parameters $[sn, tn]_{q^m}$ is an $\mathbb{F}_{q^m}$ -linear code which admits a generator matrix made of $s \times t$ ideal matrix blocks. A crucial point is that if $P \in \mathbb{F}_q[X]$ is irreducible and if n and m are prime, then this code admits a systematic generator matrix made of ideal blocks [1]. In the following, we will restrict ourselves to $t = 1$.

Definition 6 (Ideal codes). Let $P(X) \in \mathbb{F}_q[X]$ be a polynomial of degree n and let $\mathbf{g}_i \in \mathbb{F}_{q^m}^n$ for $i \in \{1, \dots, s-1\}$. We call the $[sn, n]_{q^m}$ ideal code $\mathcal{C}$ of generators $(\mathbf{g}_1, \dots, \mathbf{g}_{s-1})$ the code with generator matrix $\mathbf{G} = (\mathbf{I}_n \mathcal{IM}(\mathbf{g}_1) \dots \mathcal{IM}(\mathbf{g}_{s-1})) \in \mathbb{F}_{q^m}^{n \times sn}$. Equivalently, the code $\mathcal{C}$ admits a parity-check matrix of the form

$$\mathbf{H} = \begin{pmatrix} \mathcal{IM}(\mathbf{h}_1) \\ \vdots \\ \mathcal{IM}(\mathbf{h}_{s-1}) \end{pmatrix}.$$

Definition 7 (IRSD Problem). Given $\mathbf{H} \in \mathbb{F}_{q^m}^{(s-1)n \times sn}$ a parity-check matrix of an $[sn, n]_{q^m}$ -ideal code and $\mathbf{s} \in \times \mathbb{F}_{q^m}^{(s-1)n}$, the Ideal Rank Support Decoding Problem $\text{IRSD}(n, s, r)$ asks to compute $\mathbf{e} \in \mathbb{F}_{q^m}^{ns}$ such that $\|\mathbf{e}\| \leq r$ and $\mathbf{H}\mathbf{e}^\top = \mathbf{s}^\top$.

Definition 8 (IRSL Problem). Given $\mathbf{H} \in \mathbb{F}_{q^m}^{(s-1)n \times sn}$ a parity-check matrix of an $[sn, n]_{q^m}$ -ideal code and $\mathbf{S} \in \times \mathbb{F}_{q^m}^{N \times (s-1)n}$, the Ideal Rank Support Learning Problem $\text{IRSL}(n, s, r, N)$ asks to compute a subspace E of $\mathbb{F}_{q^m}$ of dimension r for which there exists a matrix $\mathbf{V} \in E^{N \times n}$ such that $\mathbf{H}\mathbf{V}^\top = \mathbf{S}^\top$.

2.4 LRPC codes and early LRPC-based schemes

LRPC codes were introduced in [22] as the rank metric analogue of LDPC codes.

Definition 9 (LRPC code). *An $[n, k]_{q^m}$ -linear code $\mathcal{C}$ is said to be LRPC of dual weight d if it admits a parity-check matrix $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times n}$ whose coefficients span an $\mathbb{F}_q$ -vector space F of dimension d . Such a matrix $\mathbf{H}$ will be called a homogeneous matrix of weight d and support F .*

Introduced in [22], the Rank Support Recovery (RSR) algorithm allows to decode efficiently if the support F of an homogeneous parity-check matrix is known. The following definition combines Definition 6 and Definition 9, as we can clearly construct codes which admit the two properties:

Definition 10 (Ideal-LRPC code). *An Ideal-LRPC code is both an Ideal code and an LRPC code.*

Presented in Figure 1, the LOCKER Public Key Encryption scheme [7] uses such an Ideal-LRPC code. Its security relies on the difficulty of the IRSD problem.

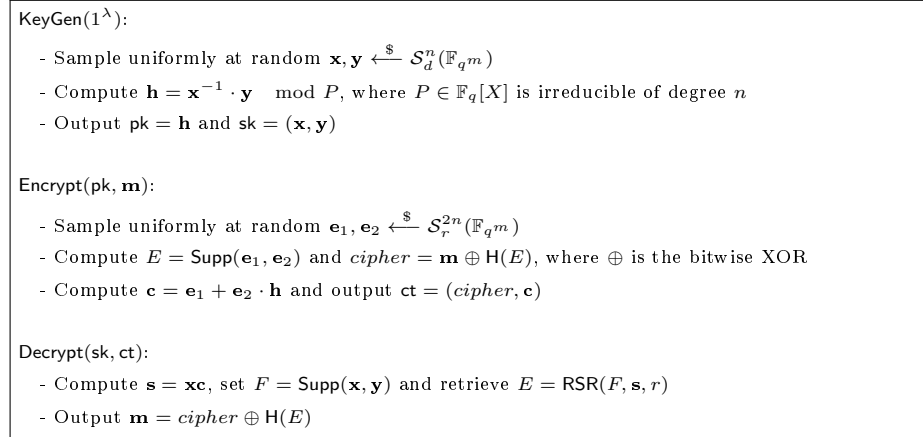


Fig. 1: Description of the LOCKER scheme

The following Key Encapsulation Mechanism (KEM) given in Figure 2 is due to [3]. It exploits several syndromes whose errors have the same support in order to improve the initial LRPC decoder. Its security relies on the IRSL problem.

2.5 Augmented Gabidulin codes and the RQC-MS-AG scheme

Augmented Gabidulin codes were introduced in [18]. The idea is to add a sequence of zeros at the end of a Gabidulin code.

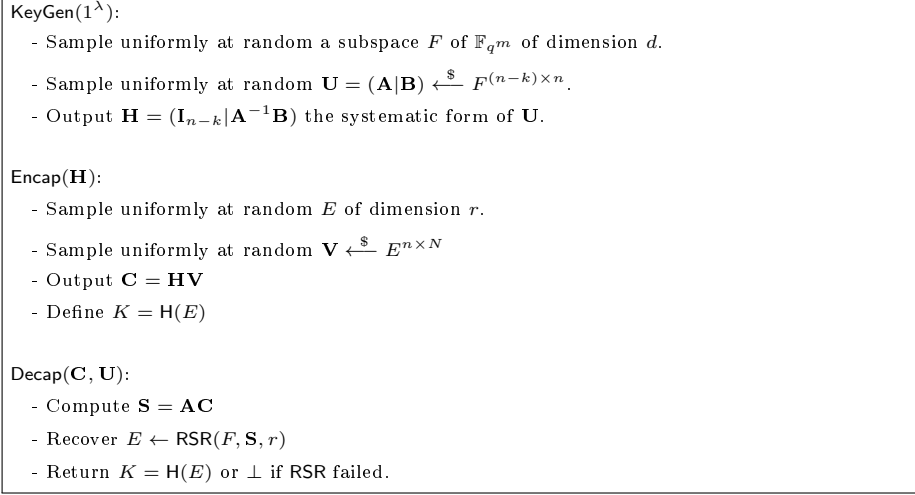


Fig. 2: Algorithms of the Key Encapsulation Mechanism ILRPC-MS

Definition 11 (Augmented Gabidulin codes). Let $(k, n, n', m) \in \mathbb{N}^4$ such that $k \leq n' < m < n$. Let $\mathbf{g} = (g_1, \dots, g_{n'}) \in \mathbb{F}_{q^m}^{n'}$ such that $\|\mathbf{g}\| = n'$ and let $\bar{\mathbf{g}} \stackrel{\text{def}}{=} (\mathbf{g} | \mathbf{0}_{n-n'}) \in \mathbb{F}_{q^m}^n$. The Augmented Gabidulin code $\mathcal{G}_{\bar{\mathbf{g}}}^+(n, n', k, m)$ is the code of parameters $[n, k]_{q^m}$ defined by:

$$\mathcal{G}_{\bar{\mathbf{g}}}^+(n, n', k, m) \stackrel{\text{def}}{=} \{P(\bar{\mathbf{g}}), \deg_q(P) < k\},$$

where $P(\bar{\mathbf{g}}) \stackrel{\text{def}}{=} (P(g_1), \dots, P(g_{n'}), \mathbf{0}_{n-n'})$ and P is a q -polynomial.

The idea is to benefit from elements of the support of the error in the last positions when we decode. They correspond to *support erasures* in a rank metric context. More precisely, *support erasures* are defined as a subspace of the vector space spanned by the error coordinates, i.e., the support of the error. Overall, these codes allow to improve the decoding capacity $\left\lfloor \frac{n'-k}{2} \right\rfloor$ of the original Gabidulin code but this comes at the price of a non-zero decryption failure rate.

Proposition 1 (Decoding Algorithm for Augmented Gabidulin codes). Let $\mathcal{G}_{\bar{\mathbf{g}}}^+(n, n', k, m)$ be an augmented Gabidulin code and let $\varepsilon \in \{1, 2, \dots, \min(n - n', n' - k)\}$ be the dimension of the vector space generated by the support erasures. There exists an efficient decoding algorithm correcting errors of rank weight up to $\delta \stackrel{\text{def}}{=} \left\lfloor \frac{n'-k+\varepsilon}{2} \right\rfloor$ with a decryption failure rate (DFR) of:

$$\text{DFR}(n, n', \delta, \varepsilon) = q^{\delta(n'-n)} \sum_{i=1}^{\varepsilon-1} \prod_{j=0}^{i-1} \frac{(q^\delta - q^j)(q^{n-n'} - q^j)}{q^i - q^j}.$$

Using such codes together with the multi syndrome approach of [3] allowed to devise an improvement of RQC called RQC-MS-AG [18]. This scheme is declined in two versions. What is important for our purposes is that one uses *non-homogeneous* errors. A non-homogeneous vector of weight (ω_1, ω_2) in $\mathbb{F}_{q^m}^{3n}$ is an element of

$$\mathcal{S}_{(\omega_1, \omega_2)}^{3n}(\mathbb{F}_{q^m}) \stackrel{\text{def}}{=} \{ \mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \in \mathbb{F}_{q^m}^{3n} \mid \|(\mathbf{x}_1, \mathbf{x}_3)\| = \omega_1, \\ \|\mathbf{x}_2\| = \omega_1 + \omega_2, \text{Supp}(\mathbf{x}_1, \mathbf{x}_3) \subset \text{Supp}(\mathbf{x}_2) \}.$$

The use of several syndromes requires to extend this notion to matrices (the support still corresponding to the vector space spanned by its coefficients):

$$\mathcal{S}_{(\omega_1, \omega_2)}^{N \times 3n}(\mathbb{F}_{q^m}) \stackrel{\text{def}}{=} \{ \mathbf{M} = (\mathbf{M}_1 \mid \mathbf{M}_2 \mid \mathbf{M}_3) \in \mathbb{F}_{q^m}^{N \times 3n}, \dim(\text{Supp}(\mathbf{M}_1 \mid \mathbf{M}_3)) = \omega_1, \\ \dim(\text{Supp}(\mathbf{M}_2)) = \omega_1 + \omega_2, \text{Supp}(\mathbf{M}_1 \mid \mathbf{M}_3) \subset \text{Supp}(\mathbf{M}_2) \}.$$

Figure 3 presents the RQC-MS-AG scheme using non-homogeneous errors. As it also uses ideal codes, we consider n_1 and n_2 two integers and $P \in \mathbb{F}_q[X]$ an irreducible polynomial of degree n_2 . For a vector $\mathbf{v} \in \mathbb{F}_{q^{n_2}}^{n_1}$ and a matrix $\mathbf{M} \in \mathbb{F}_{q^m}^{n_2 \times n_1}$, we generalize the internal product between vectors by

$$\mathbf{v} \cdot \mathbf{M} \stackrel{\text{def}}{=} ((\mathbf{v} \cdot \mathbf{m}_1)^\top, \dots, (\mathbf{v} \cdot \mathbf{m}_{n_1})^\top),$$

where $\mathbf{m}_i$ is the i -th column of $\mathbf{M}$ for $i \in \{1, \dots, n_1\}$ and where the products at the right hand side are standard internal products. The procedure **Fold** turns the vector $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_{n_1}) \in (\mathbb{F}_{q^{n_2}}^{n_1})$ into $\text{Fold}(\mathbf{v}) \stackrel{\text{def}}{=} (\mathbf{v}_1^\top, \dots, \mathbf{v}_{n_1}^\top) \in \mathbb{F}_{q^m}^{n_2 \times n_1}$. The inverse map is denoted by **Unfold**.

KeyGen(1^λ):

- Sample uniformly at random $\mathbf{h} \xleftarrow{\$} \mathbb{F}_{q^m}^{n_2}$, $\mathbf{g} \xleftarrow{\$} \mathcal{S}_{n'}^{n'}(\mathbb{F}_{q^m})$ and $(\mathbf{x}, \mathbf{y}) \xleftarrow{\$} \mathcal{S}_{\omega, 1}^{2n_2}(\mathbb{F}_{q^m})$.
- Compute $\mathbf{s} = \mathbf{x} + \mathbf{h} \cdot \mathbf{y} \pmod{P}$
- Output $\text{pk} = (\mathbf{g}, \mathbf{h}, \mathbf{s})$ and $\text{sk} = (\mathbf{x}, \mathbf{y})$

Encrypt($\text{pk}, \mathbf{m}$):

- Compute a generator matrix $\mathbf{G} \in \mathbb{F}_{q^m}^{k \times n_1 n_2}$ for $\mathcal{G}_{\mathbf{g}}^+(n_1 n_2, n', k, m)$, $\bar{\mathbf{g}} \stackrel{\text{def}}{=} (\mathbf{g} \mid \mathbf{0}_{n_1 n_2 - n'})$
- Sample uniformly at random $(\mathbf{R}_1, \mathbf{E}, \mathbf{R}_2) \xleftarrow{\$} \mathcal{S}_{(\omega_1, \omega_2)}^{n_2 \times 3n_1}(\mathbb{F}_{q^m})$
- Compute $\mathbf{U} = \mathbf{R}_1 + \mathbf{h} \cdot \mathbf{R}_2$ and $\mathbf{V} = \text{Fold}(\mathbf{mG}) + \mathbf{s} \cdot \mathbf{R}_2 + \mathbf{E}$
- Output $\mathbf{C} = (\mathbf{U}, \mathbf{V})$

Decrypt($\text{sk}, \mathbf{C}$):

- Output $\mathcal{G}_{\bar{\mathbf{g}}}^+ \cdot \text{Decode}(\text{Unfold}(\mathbf{V} - \mathbf{y} \cdot \mathbf{U}))$

Fig. 3: Description of the RQC-MS-AG scheme

2.6 Blockwise errors and related problems

Blockwise errors have been recently introduced in [31]. Their particular structure was used to increase the capacity of LRPC decoding.

Definition 12 (Blockwise ℓ -error). Let $\mathbf{n} = (n_1, \dots, n_\ell) \in \mathbb{N}^\ell$, $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{N}^\ell$ and $n \stackrel{\text{def}}{=} \sum_{i=1}^\ell n_i$. An error $\mathbf{e} \in \mathbb{F}_{q^m}^n$ is said to be an ℓ -error with parameters $\mathbf{n}$ and $\mathbf{r}$ if it is the concatenation of ℓ errors $\mathbf{e}_i \in \mathbb{F}_{q^m}^{n_i}$ such that

- for all $i \in \{1, \dots, \ell\}$, $\|\mathbf{e}_i\| = r_i$,
- for all $i \neq j$, $\text{Supp}(\mathbf{e}_i) \cap \text{Supp}(\mathbf{e}_j) = \{0\}$.

We denote $\mathcal{S}_{\mathbf{r}}^{\mathbf{n}}(\mathbb{F}_{q^m})$ as the set of blockwise errors with parameters $\mathbf{n}$ and $\mathbf{r}$. For an integer N and vectors $\mathbf{n}$ and $\mathbf{r}$, we can similarly define $\mathcal{S}_{\mathbf{r}}^{N \times \mathbf{n}}(\mathbb{F}_{q^m})$ the set of matrices of size $N \times n_i$ whose elements are block matrices $\mathbf{M} = (\mathbf{M}_1 \mid \dots \mid \mathbf{M}_\ell)$ such that $\dim(\text{Supp}(\mathbf{M}_i)) = r_i$. We can naturally define restrictions of the RD and IRSD problems to blockwise errors.

Definition 13 (ℓ -RD problem). Let $\mathbf{n} = (n_1, \dots, n_\ell) \in \mathbb{N}^\ell$, $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{N}^\ell$ and $n \stackrel{\text{def}}{=} \sum_{i=1}^\ell n_i$. Given a full-rank matrix $\mathbf{G} \in \mathbb{F}_{q^m}^{k \times n}$ and $\mathbf{y} \stackrel{\text{def}}{=} \mathbf{x}\mathbf{G} + \mathbf{e}$ such that $\mathbf{x} \in \mathbb{F}_{q^m}^k$ is uniformly sampled and $\mathbf{e} \in \mathcal{S}_{\mathbf{r}}^{\mathbf{n}}$, the Blockwise Rank Decoding problem $\text{RD}(\mathbf{n}, k, \mathbf{r}, m)$ asks to find $\mathbf{x}$ and $\mathbf{e}$.

Definition 14 (ℓ -IRSD problem). Let $\mathbf{n} = (n_1, \dots, n_\ell) \in \mathbb{N}^\ell$, $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{N}^\ell$ and $n \stackrel{\text{def}}{=} \sum_{i=1}^\ell n_i$. Let $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-1)s \times ns}$ be a parity-check matrix of an $[sn, n]$ ideal code. On input $(\mathbf{H}, \mathbf{s})$ where $\mathbf{s}^\top = \mathbf{H}\mathbf{e}^\top$ and $\mathbf{e} \in \mathcal{S}_{\mathbf{r}}^{\mathbf{n}}$, the Blockwise Ideal Rank Syndrome Decoding problem $\text{IRSD}(\mathbf{n}, k, \mathbf{r}, m)$ asks to find $\mathbf{e}$.

An improved version of LOCKER based on 2-IRSD was given in [31].

3 ℓ -LRPC codes and decoding with several syndromes

In this paper, we combine the multi syndrome approach of [3] together with the blockwise structure of [31]. Thus, Section 3.1 starts by describing new restrictions of RSL to this error structure.

3.1 New problems related to blockwise errors

Definition 15 (ℓ -RSL problem). Given $(\mathbf{H}, \mathbf{H}\mathbf{E}^\top)$, where $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times n}$ is full-rank and where $\mathbf{E} = (\mathbf{E}_1 \mid \dots \mid \mathbf{E}_\ell) \in \mathbb{F}_{q^m}^{N \times n}$ is such that for $i \in \{1, \dots, \ell\}$, the matrix $\mathbf{E}_i \in \mathbb{F}_{q^m}^{N \times n_i}$ is homogeneous of support $\mathcal{V}_i$, $\dim \mathcal{V}_i = r_i$, $\mathcal{V}_i \cap \mathcal{V}_j = \{0\}$ for $i \neq j$, the Blockwise Rank Support Learning problem $\ell\text{-RSL}(m, \mathbf{n}, \mathbf{r}, k, N)$ asks to find the set of subspaces $(\mathcal{V}_i)_{i \in \{1, \dots, \ell\}}$.

We can also define a variant of this problem for an ideal code of parameters $[sn, n]_{q^m}$ and where the s -errors have blocks of the same length n .

Definition 16 (s -IRSL problem). Let $\mathbf{H}$ be a parity check matrix of an $[sn, n]_{q^m}$ ideal code and let $\mathbf{r} = (r_1, \dots, r_s) \in \mathbb{N}^s$. Given $(\mathbf{H}, \mathbf{S}) \in \mathbb{F}_{q^m}^{(s-1)n \times sn} \times \mathbb{F}_{q^m}^{N \times (s-1)n}$, the Blockwise Ideal Rank Support Learning problem $\text{IRSL}(s, n, \mathbf{r}, N)$ asks to compute a set of s subspaces $\mathcal{V} = (\mathcal{V}_1, \dots, \mathcal{V}_s)$ such that $\dim \mathcal{V}_i = r_i$, $\mathcal{V}_i \cap \mathcal{V}_j = \{0\}$ for $i \neq j$ and such that there exists a matrix $\mathbf{V} = (\mathbf{V}_1 \mid \dots \mid \mathbf{V}_s) \in \mathbb{F}_{q^m}^{N \times sn}$ such that $\mathbf{H}\mathbf{V}^\top = \mathbf{S}^\top$ and whose i -th block is homogeneous of support $\mathcal{V}_i$ for all $i \in \{1..s\}$.

In the rest of the section, we study decoding algorithms for ℓ -LRPC codes, introduced in [31]. Their definition is recalled below.

Definition 17. Let $\mathbf{H} = (\mathbf{H}_1 \mid \dots \mid \mathbf{H}_\ell) \in \mathbb{F}_{q^m}^{(n-k) \times n}$ full-rank such that $\mathbf{H}_i \in \mathbb{F}_{q^m}^{(n-k) \times n_i}$ is homogeneous of weight d_i and support F_i for $i \in \{1..\ell\}$ and such that for all $i \neq j$, $F_i \cap F_j = \{0\}$. The code $\mathcal{C}$ with parity-check matrix $\mathbf{H}$ is said to be an ℓ -LRPC code (with dual weight $(d_1, \dots, d_\ell)$).

In Section 3.2, we extend the decoding algorithm of [31] to multiple syndromes. In Section 3.3, we propose a way to improve its DFR by using a trick from [3].

3.2 Decoding algorithm with multiple syndromes

Our new algorithm is described in Algorithm 1. Its correctness easily follows from the one of the algorithms of [10, 31].

Algorithm 1 Decoding algorithm of ℓ -LRPC codes for ℓ -errors

Input: A collection of N syndromes $(\mathbf{s}_1, \dots, \mathbf{s}_N) \in \mathbb{F}_{q^m}^{(n-k) \times N}$ and the parity-check matrix $\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times k}$

Output: The ℓ -error $\mathbf{e}$, or error

 Compute the syndrome space $S = \langle s_{1,1}, \dots, s_{N,n-k} \rangle$

 Let $\{F_{i1}, \dots, F_{id_i}\}$ be a basis of F_i for all i

 Compute $S_{ij} = F_{ij}^{-1}S$ for all $i \in \{1, \dots, \ell\}$ and $j \in \{1, \dots, d_i\}$

 Compute $E_i = \bigcap_{j=1}^{d_i} S_{ij}$

if $\dim(E_i) \neq r_i$ for any i **then**

return error

else

 Recover $E = \sum_{i=1}^{\ell} E_i$

 Solve the linear system $\mathbf{H}\mathbf{e}^\top = \mathbf{s}^\top$ with $\mathbf{e} \in E^n$ as unknown

return e

This algorithm has a non-zero DFR. There are two cases that can make it fail:

1. the dimension of the syndrome space S is lower than the dimension of the whole product space $\sum_{i=1}^{\ell} E_i F_i$;
2. there exists $i \in \{1..\ell\}$ such that $E_i \not\supseteq \bigcap_{j=1}^{d_i} S_{ij}$.

An upper bound of this DFR is given in Theorem 1.

Theorem 1. *Let $\mu = \sum_{i=1}^{\ell} r_i d_i$ and let N be the number of syndromes. Under the assumption that each element of the syndrome space as a random element of $P \stackrel{\text{def}}{=} \sum_{i=1}^{\ell} E_i F_i$, the decoding failure probability of Algorithm 1 is bounded by:*

$$q^{-(N(n-k)-\mu)} + \sum_{i=1}^{\ell} q^{-(d_i-1)(m-\mu)+\mu-r_i}. \quad (1)$$

To prove it, we need the following result from [7]:

Proposition 2. *Let r, d and μ be three integers. Let E be a fixed subspace of dimension r and let $R_i, 1 \leq i \leq d$, be d independently chosen random subspaces of dimension μ containing the subspace E . The probability that $\dim \bigcap_{i=1}^d R_i > r$ is bounded from above by:*

$$q^{\mu-r} \left(\frac{q^{\mu} - q^r}{q^{\mu}} \right)^{d-1} \approx q^{-(d-1)(m-\mu)+\mu-r}$$

Proof (of Theorem 1). First, we study the probability that $\dim(S) < \dim(\sum E_i F_i)$. Each s_{ij} is an element of the product space $P = \sum E_i F_i$. Thus, we can write the set of coefficients of all syndromes as an element in $\mathbb{F}_q^{N(n-k) \times \mu}$ whose rows are obtained by unfolding the s_{ij} 's in a fixed basis of P . By assumption, this matrix behaves as a random matrix. Under this assumption, the probability that $\dim(S) < \dim(P)$ is thus equal to the probability that a random $N(n-k) \times \mu$ matrix is not full-rank. This probability can be upper-bounded by $q^{-(N(n-k)-\mu)}$ and this gives the first term in Equation (1). The second case which leads to a decoding failure is when there is $i \in \{1..\ell\}$ such that $E_i \not\supseteq \bigcap_{j=1}^{d_i} S_{ij}$. By Proposition 2, the probability that $E_i \not\supseteq \bigcap_{j=1}^{d_i} S_{ij}$ can be upper bounded by $q^{-(d_i-1)(m-\mu)+\mu-r_i}$ for $i \in \{1..\ell\}$. We need to recover E_i for all $1 \leq i \leq \ell$, hence the result. $\square$

3.3 Improving its DFR

By using a technique introduced in the xMS protocol [3], we extend Algorithm 1 to reduce its DFR. The resulting algorithm corresponds to Algorithm 2.

Algorithm 2 Decoding algorithm of ℓ -LRPC codes for ℓ -errors

Input: A collection of N syndromes $(\mathbf{s}_1, \dots, \mathbf{s}_N) \in \mathbb{F}_{q^m}^{(n-k) \times N}$, the parity-check matrix

$\mathbf{H} \in \mathbb{F}_{q^m}^{(n-k) \times k}$ and an algorithm parameter c

Output: The ℓ -error $\mathbf{e}$, or **error**

Compute the syndrome space $S = \langle s_{1,1}, \dots, s_{N,n-k} \rangle$

Let $\{F_{i1}, \dots, F_{id_i}\}$ be a basis of F_i for all i

Compute $S_{ij} = F_{ij}^{-1}S$ for all $i \in \{1, \dots, \ell\}$ and $j \in \{1, \dots, d_i\}$

Compute $E_i = \bigcap_{j=1}^{d_i} S_{ij}$

if $\dim(E_i) > r_i + c$ **for any** i **then**

return error

else

$E' = \sum_{i=1}^{\ell} E_i$

 Solve the linear system $\mathbf{H}\mathbf{e}^\top = \mathbf{s}^\top$ with $\mathbf{e} \in E'^n$ as unknown

return e

Correctness. The parameter c must be chosen so that the linear system over $\mathbb{F}_q$ derived from $\mathbf{H}\mathbf{e}^\top = \mathbf{s}^\top$ with the knowledge of E' has more linearly independent equations than the number of unknowns. If these equations are linearly independent, this condition is met when $(n-k)m \geq \sum n_i \dim(E_i)$, hence a fortiori when $(n-k)m \geq \sum n_i(r_i + c)$. When the system has a unique solution, the rest of the algorithm works in the same way as in Algorithm 1.

Theorem 2. Let $\mu = \sum r_i d_i$ and let N be the number of given syndromes. Under the same assumption as in Theorem 1, the decoding failure probability (DFR) of the extended decoding algorithm for ℓ -LRPC codes is bounded by:

$$q^{-(N(n-k)-\mu)} + \frac{1}{\phi(q^{-1})} \sum_{i=1}^{\ell} q^{(c+1)(\mu-r_i-(c+1)+(d_i-1)(\mu-m))},$$

where ϕ is the Euler function $\phi(x) \stackrel{\text{def}}{=} \prod_{k=1}^{+\infty} (1 - x^k)$, $|x| < 1$.

Proof. The improvement is in the second term, the first term $q^{-(N(n-k)-\mu)}$ being similar to the one of Theorem 1. Another possibility for Algorithm 2 to fail is if $\dim(E_i) > r_i + c$ for at least one $i \in \{1.. \ell\}$. By [3, Proposition 3], we have

$$P\left(\dim\left(\bigcap_{j=1}^{d_i} S_{ij}\right) > r_i + c\right) \leq \frac{1}{\phi(q^{-1})} q^{(c+1)(\mu-r_i-(c+1)+(d_i-1)(\mu-m))}.$$

As in the proof of Theorem 1, the second term follows by summing the upper bounds for $i \in \{1.. \ell\}$. $\square$

4 New cryptographic schemes based on ℓ -RSL and ℓ -IRSL

4.1 RQC-MS-AG scheme with blockwise errors

We propose an improvement of the RQC-MS-AG by using 2-errors and 3-errors. A description of the resulting scheme can be found in Figure 4.

Comments. The Augmented Gabidulin code has parameters $(n_1 n_2, m, k, m)$ and Decode is an efficient decoding algorithm that can correct up to $\delta = \lfloor \frac{m-k+\varepsilon}{2} \rfloor$ errors, where $\varepsilon \leq \min(m-k, n_1 n_2 - m)$ is fixed as a parameter (in this case, the DFR is estimated by Proposition 1). The main difference with the former RQC-MS-AG scheme is that $(\mathbf{x}, \mathbf{y})$ is a 2-blockwise error rather than a random error of length $2n_2$ whose support contains 1 and that the triple $(\mathbf{R}_1, \mathbf{E}, \mathbf{R}_2)$ sampled at the encryption is a set of 3-blockwise errors of the same support instead of being a set of non-homogeneous errors with the same support. The rest of the scheme is rather similar and we keep the same notation as in Figure 3.

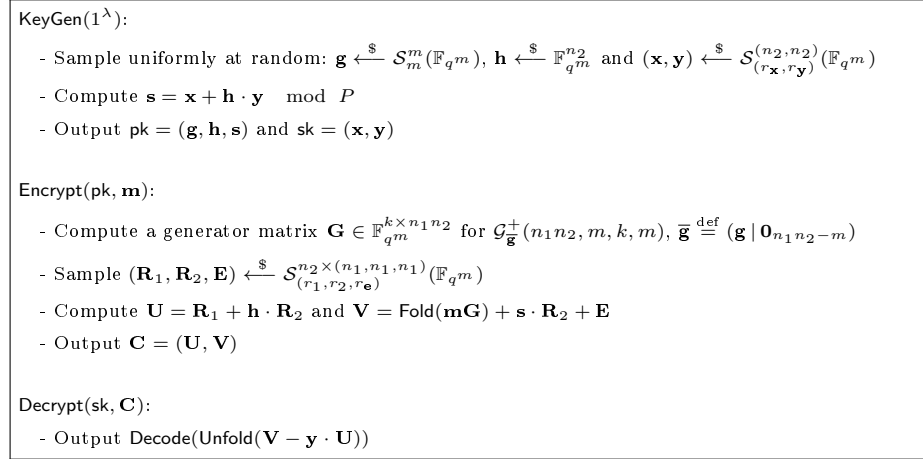


Fig. 4: Description of the RQC-MS-AG scheme with blockwise errors

The parameters need to be chosen according to the following proposition.

Proposition 3. *Decryption is correct as long as*

$$\|\text{Unfold}(\mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E})\| \leq \delta.$$

Proof. We have $\mathbf{U} = \mathbf{R}_1 + \mathbf{h} \cdot \mathbf{R}_2$ and $\mathbf{V} = \text{Fold}(\mathbf{mG}) + \mathbf{s} \cdot \mathbf{R}_2 + \mathbf{E}$, so that

$$\begin{aligned} \mathbf{V} - \mathbf{y} \cdot \mathbf{U} &= \text{Fold}(\mathbf{mG}) + (\mathbf{x} + \mathbf{h}\mathbf{y}) \cdot \mathbf{R}_2 + \mathbf{E} - \mathbf{y} \cdot (\mathbf{R}_1 + \mathbf{h}\mathbf{R}_2) \\ &= \text{Fold}(\mathbf{mG}) + \mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E}. \end{aligned}$$

This implies $\text{Unfold}(\mathbf{V} - \mathbf{y} \cdot \mathbf{U}) = \mathbf{m}\mathbf{G} + \text{Unfold}(\mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E})$. Therefore, the algorithm `Decode` will output $\mathbf{m}$ (there is still a DFR) as long as $\|\text{Unfold}(\mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E})\| \leq \delta$. $\square$

An optimization: choose the matrix $\mathbf{R}_2$ with 1 in its support. Recall that the error to correct is equal to:

$$\text{Err} = \mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E}.$$

Note that if $1 \in \text{Supp}\mathbf{R}_2$, then $\text{Supp}(\mathbf{x}) \subset \text{Supp}(\text{Err})$. By adding this constraint for the sampling of the vectors, we can deduce a subset of dimension r_x of the support of the error to correct. In order to find the remaining coordinates, the support erasures (which span a space of dimension $\delta - r_x$ after project the coordinates of the error in the space $\text{Supp}(\mathbf{x})^\perp$) have to span a space of dimension $\varepsilon - r_x$. We deduce that the decoding failure rate knowing the support of $\mathbf{x}$ is given by:

$$\text{DFR}(n, n', \varepsilon - r_x, \delta - r_x)$$

where DFR is the formula given in proposition 1.

We can impose to the support of the block vectors $(\mathbf{R}_1, \mathbf{R}_2, \mathbf{E})$ to contain 1 in one of the blocks without changing the practical complexity of the attacks (these block errors are the solution of the 3-IRSL problem on the $[3n_2, n_2]_{q^m}$ ideal code whose $\begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{h} \\ \mathbf{0} & \mathbf{1} & \mathbf{s} \end{pmatrix}$ is a parity check matrix).

Indeed, let us consider an instance $(\mathbf{H}, \mathbf{s}) \in \mathbb{F}_{q^m}^{(n-k) \times n} \times \mathbb{F}_{q^m}^{n-k}$ of the Rank-Syndrome-Decoding problem, with $\mathbf{H}$ the parity check matrix of a code $\mathcal{C}$. Let $\mathbf{e} \in \mathbb{F}_{q^m}^n$ such that $\mathbf{s} = \mathbf{e}\mathbf{H}^\top$ and $\|\mathbf{e}\| \leq r$. By solving this linear system without considering the constraint on the weight of $\mathbf{e}$, one can find a vector $\mathbf{y} = \mathbf{c} + \mathbf{e}$, with $\mathbf{c} \in \mathcal{C}$. We define the code: $\tilde{\mathcal{C}} = \mathcal{C} \oplus \langle \mathbf{y} \rangle$. We reduced the original problem to find a vector of rank r in the code $\tilde{\mathcal{C}}$. One must solve the instance of the Rank Syndrome Decoding problem:

$$\mathbf{e}'\tilde{\mathbf{H}}^\top = 0$$

whose the set of solutions is $\{\lambda\mathbf{e} \mid \lambda \in \mathbb{F}_{q^m}^*\}$.

The best known attacks on the Rank-Syndrome-Decoding problem use the reduction above ([28], [11], [15]). Therefore, we consider in practice the best attacks on the Rank-Syndrome-Decoding problem to evaluate the security of an instance whose we know that 1 belongs to the support of the error.

A similar reasoning can be made in the case of blockwise errors and multiple syndromes, and we can choose an error with 1 in the desired block. In our case, we can take advantage of the protocol if we choose to sample a set of n_1 errors: $(\mathbf{R}_1, \mathbf{R}_2, \mathbf{E}) \in \mathcal{S}_{(r_1, r_2, r_e)}^{n_2 \times (n_1, n_1, n_1)}(\mathbb{F}_{q^m})$ with 1 in the support of $\mathbf{R}_2$.

4.2 ILRPC-MS with blockwise errors

We also improve the ILRPC-MS scheme of [3] described in Figure 2 by using 2-errors. Our new scheme is presented in Figure 5.

Let $\mathcal{V} = (\mathcal{V}_i)_{i \in \{1, \dots, \ell\}}$ a finite sequence of subspaces of $\mathbb{F}_{q^m}$ such that $\dim \mathcal{V}_i = r_i$ and for all $i \neq j$: $\mathcal{V}_i \cap \mathcal{V}_j = \{0\}$. We denote $\mathcal{S}_{\mathbf{r}}^{\mathbf{n}}(\mathcal{V})$ the set of vectors of the form $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_\ell)$, such that for all $i \in \{1, \dots, \ell\}$, the coefficients of each vector $\mathbf{x}_i \in \mathbb{F}_{q^m}^{n_i}$ belongs to $\mathcal{V}_i$.

KeyGen(1^λ):

- Choose uniformly at random two subspaces F_1 and F_2 in $\mathbb{F}_{q^m}$ of respective dimensions d_1 and d_2 .
- Sample a couple of polynomials whose coefficients belong to F : $(\mathbf{x}, \mathbf{y}) \xleftarrow{\$} F_1^{n_2} \times F_2^{n_2}$.
- Compute $\mathbf{h} = \mathbf{x}^{-1} \mathbf{y} \mod P$
- Output $\mathbf{pk} = \mathbf{h}$ and $\mathbf{sk} = (\mathbf{x}, \mathbf{y})$

Encap($\mathbf{pk}$):

- Choose uniformly at random $\mathcal{V} = (\mathcal{V}_1, \mathcal{V}_2)$ such that $\dim \mathcal{V}_i = r_i$ and $\mathcal{V}_1 \cap \mathcal{V}_2 = \{0\}$
- Sample uniformly n_1 polynomials whose coefficients belong to $\mathcal{S}_{\mathbf{r}}^{(n_2, n_2)}(\mathcal{V})$:

$(\mathbf{e}_1, \dots, \mathbf{e}_{n_1}) \xleftarrow{\$} (\mathcal{S}_{\mathbf{r}}^{(n_2, n_2)}(\mathcal{V}))^{n_1}$

- Write each vector $\mathbf{e}_i$ as concatenation of $\mathbf{e}_{i,1}$ and $\mathbf{e}_{i,2}$, i.e. $\mathbf{e}_i = (\mathbf{e}_{i,1} | \mathbf{e}_{i,2})$
- Compute $\mathbf{c}_i = \mathbf{e}_{i,1} + \mathbf{e}_{i,2} \mathbf{h}$ for all integer i from 1 to n_1 .
- Define $K = H(\mathcal{V})$ and output $\mathbf{c} = (\mathbf{c}_1, \dots, \mathbf{c}_{n_1})$

Decap($\mathbf{sk}, \mathbf{c}$):

- Compute $\mathbf{S} = (\mathbf{x}\mathbf{c}_1, \dots, \mathbf{x}\mathbf{c}_{n_1})$
- Recover $\mathcal{V} \leftarrow \text{Decode}(F, \mathbf{S}, \mathbf{r})$
- Return $K = H(\mathcal{V})$ or $\perp$ if the **Decode** algorithm failed.

Fig. 5: Algorithms **KeyGen**, **Encap** and **Decap** of the Key Encapsulation Mechanism ILRPC-Block-MS

Comments. As ideal codes are used, we recall that the vectors $\mathbf{x}, \mathbf{y}$ in this figure must be seen as elements in $\mathbb{F}_{q^m}[X]$ taken modulo an irreducible polynomial $P \in \mathbb{F}_q[X]$ of degree n . The **Decode** algorithm is a decoding algorithm for LRPC codes in the case of blockwise errors. It can be either Algorithm 1 or Algorithm 2. More precisely, we call our scheme ILRPC-Block-MS when Algorithm 1 is used and ILRPC-Block-XMS($r + c$) otherwise, where c is the extra parameter in Algorithm 2. These two algorithms output the error vector rather than its support but they are somehow equivalent to RSR because it is straightforward to recover the full error vector once its support is known.

5 Combinatorial attacks

In this section, we present combinatorial attacks against three difficult problems adapted to blockwise errors:

1. For the ℓ – RD problem, we present an adaptation of the AGHT attack, different from [31], as well as a new attack called *Shortening and Truncating*. We compare these attacks on a specific parameter case;
2. For the ℓ – RSL problem
3. A structural attack against ℓ -LRPC codes.

5.1 Combinatorial attacks against ℓ -RD

To study the complexity of solving the ℓ -RD problem with combinatorial attacks, we will adapt and derive the new complexity of the attacks from [11, 23, 28] to the case of ℓ -errors. In this section, we present results in a simplified situation where $n_1 = \dots = n_\ell = n$, $k = n$ and $r_1 \leq r_2 \leq \dots \leq r_\ell$.

These attacks are similar to what was presented in [30], although it does not require the support to be disjoint. Another difference is that we take advantage of simplified situations as explained in the previous paragraph.

5.1.1 The Ourivski-Johansson attack As presented in [31], the complexity of the OJ attack is

$$\mathcal{O}((m(r-1) + (n-r_1))^{\omega} q^{(r_1-1)(n-r_1)+r_\ell}).$$

5.1.2 The AGHT attack

In order to adapt the algorithm from [11] to the case of ℓ -errors, we will sample ℓ different vector spaces F_i of dimension t_i , and the algorithm will succeed if $\exists \alpha$ such that $\forall i, \alpha E_i \subset F_i$. Using the same techniques as in [11] this probability can be approximated by:

$$\frac{q^m - 1}{q - 1} \prod_{i=1}^{\ell} q^{-r_i(m-t_i)}$$

Which gives a total complexity of:

$$\mathcal{O}((n-k)^3 m^3 q^{-m + \sum_{i=1}^{\ell} r_i(m-t_i)}) \quad (2)$$

Recall that we restrict ourselves to the case where $\forall i, n_i = \frac{n}{\ell}$.

The total complexity depends on the choice of t_i s. First we must choose these values such that $\sum_{i=1}^{\ell} t_i n_i \leq m - \lceil \frac{m(k+1)}{n} \rceil$ for the system to have more equations than unknowns, and $t_i > r_i$ for having a non-zero probability that $E_i \subset F_i$. Then there are two cases:

1. All of the r_i s are equal. In this case the choice of the t_i s does not change the complexity, and the complexity is the same for ℓ -errors and an error of weight r .
2. The r_i s are not equal. In this case the optimal strategy is to try to make perfect guesses for the smaller r_i s (i.e choosing $t_i = r_i$) in order to have the highest possible value for the t_i corresponding to the highest r_i .

The more the r_i s are different, the bigger the advantage of specifically targeting ℓ -errors instead of errors of weight r .

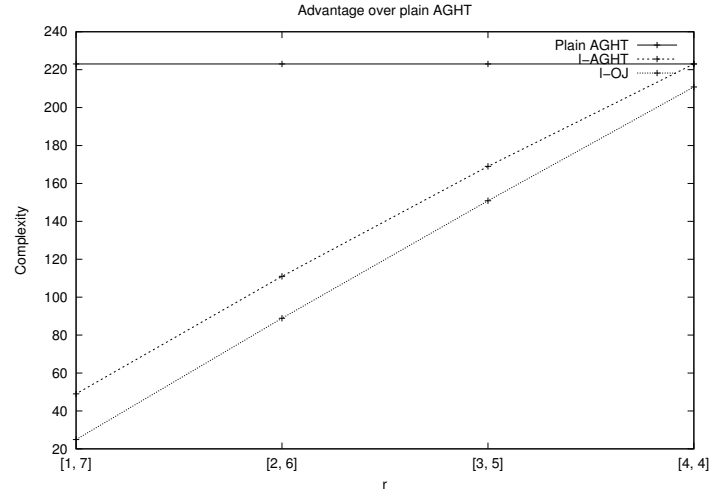


Fig. 6: Complexities of the AGHT algorithm targeting an error of rank r (plain) and adapted to ℓ -errors for parameters $m = 61, n = 134, k = 67$ and different values of $\mathbf{r}$.

Comparison with [31].

In [31, Section 3.3], the authors propose an adaptation of the AGHT attack to the case of ℓ -errors. We claim their adaptation misestimates the complexity of ℓ -AGHT attack. We give below two arguments to support our assertion.

First, in the demonstration of their Lemma 3.5 (cf. [31, Appendix C.1]), they seem to imply that the number of subspaces of $\mathbb{F}_{q^m}$ of dimension t_2 disjoint from a fixed E_1 is exactly equal to the number of subspaces of $\mathbb{F}_{q^m}/E_1$ of dimension t_1 , which is not the case. In particular, in their $\ell = 2$ example, they guess a subspace F_2 in $\mathbb{F}_{q^m}/E_1$, but in order to perform the rest of the attack, this F_2 needs to be lifted in $\mathbb{F}_{q^m}$ into a $\hat{F}_2$. Even though F_2 contains E_2/E_1 , it is not guaranteed that $\hat{F}_2$ will contain E_2 , as it depends on the choice of the representatives for the lifting.

Second, as we understood their attack, sampling F_ℓ requires a correct guess for each $E_1, \dots, E_{\ell-1}$. Therefore $F_1, \dots, F_{\ell-1}$ play no role in the attack, which sounds somewhat strange.

5.1.3 Hybrid shortening and truncating attack

This new attack is an hybrid between Ourivski-Johansson and other attacks against the plain RD problem. The attack consists of reducing the problem to solving the same problem in a code with smaller dimension (shortening), and then considering only the part of the code associated to error coordinates belonging to vectorial space of dimension r_1 (truncating). Then, we obtain a Rank Decoding problem instance with a homogeneous error of smaller dimension. It is related to the hybrid attack presented in [15, Section 5.5], with the difference that the truncating part was previously unpublished.

To simplify the analysis, let us present an attack of the 2-RD problem in a code $\mathcal{C}$ of size $[2n, n]$: let $\mathbf{G} \in \mathbb{F}_{q^m}^{n \times 2n}$ the generator matrix of $\mathcal{C}$, an error $\mathbf{e} \in S_{(r_1, r_2)}^{(n, n)}$ with $(r_1, r_2) \in \mathbb{N}^2$. We reduce the problem to the resolution of a homogeneous RD problem, in a code with smaller parameters. Let $\mathbf{y} = \mathbf{x}\mathbf{G} + \mathbf{e}$ with $\mathbf{x} \in \mathbb{F}_{q^m}^n$.

We can perform $\mathbb{F}_q$ -linear combinations on coordinates of $\mathbf{e}_1$, in order to obtain 0 in the first t_1 coordinates. In other words, it is possible to apply a matrix $\mathbf{P}$ with $r_1 t_1$ unknowns in $\mathbb{F}_q$ such that $\mathbf{e}\mathbf{P}$ is $(0 \dots 0 \mid \mathbf{e}'_1 \mid \mathbf{e}_2)$.

The attacker can then apply the same operations on the syndrome, and gets

$$\mathbf{y}' = \mathbf{y}\mathbf{P} = \mathbf{x}\mathbf{G}' + \mathbf{e}'$$

with $\mathbf{G}' = \mathbf{G}\mathbf{P}$. Without loss of generality, the matrix $\mathbf{G}$ can be in a semi-systematic form

$$\mathbf{G}' = \left(\begin{array}{c|c} I_t & * \\ \hline 0 & * \end{array} \right)$$

Operations on the columns can then be performed to cancel to top-right block of $\mathbf{G}'$, i.e. there exists an invertible matrix $\mathbf{Q}$ such that

$$\mathbf{G}'\mathbf{Q} = \left(\begin{array}{c|c} I_t & 0 \\ \hline 0 & \mathbf{A} \end{array} \right)$$

Because the error $\mathbf{e}'$ has its first t coordinates set to 0, $\mathbf{e}'\mathbf{Q} = \mathbf{e}'$ hence by writing:

$$\begin{aligned} \mathbf{y}'', & \text{ the } n \text{ rightmost coordinates of } \mathbf{y}'\mathbf{Q} \\ \mathbf{x}'', & \text{ the } n - t \text{ rightmost coordinates of } \mathbf{x} \\ \mathbf{G}'', & \text{ the } n \text{ rightmost columns of } \mathbf{A} \end{aligned}$$

we get

$$\mathbf{y}'' = \mathbf{x}''\mathbf{G}'' + \mathbf{e}_2$$

which is an instance of the RD problem in a code of parameters $[n, n - t_1, r_2]$. The cost of transforming the initial instance in this reduced instance is $q^{r_1 t_1}$ (for finding the correct matrix $\mathbf{P}$) times n^2 (for calculating the matrix $\mathbf{Q}$).

By symmetry, another variant of the attack consists in canceling t_2 coordinates in the rightmost part of the error of weight r_2 , and then solving an RD instance in a code with parameters $[n, n - t_2, r_1]$.

In the above explanation, the attacker *truncates* until obtaining a plain RD instance. Another possibility is to truncate only $t_1 \leq u_1 < n$ columns of $\mathbf{G}''$, yielding a 2-RD instance $(n - u_1, n)$ with weights (r_1, r_2) .

We can then deduce the following proposition:

Proposition 4. *The complexity of solving the 2-RD problem in a code of size (n, n) by the Shortening and Truncating attack is estimated as:*

$$n^2 \cdot \min_{\substack{1 \leq t_1 \leq n \\ 1 \leq t_2 \leq n \\ t_1 \leq u_1 \leq n \\ t_2 \leq u_2 \leq n}} (q^{r_1 t_1} \times \mathcal{T}_{2\text{-RD}}((n - u_1, n), n - t_1, (r_1, r_2), m), q^{r_2 t_2} \times \mathcal{T}_{2\text{-RD}}((n, n - u_2), n - t_2, (r_1, r_2), m)) \quad (3)$$

where $\mathcal{T}_{2\text{-RD}}(\mathbf{n}, k, \mathbf{r}, m)$ is the complexity of the best algorithm for solving an instance of 2-RD($\mathbf{n}, k, \mathbf{r}, m$) problem.

5.2 Combinatorial attacks on ℓ -RSL

The first combinatorial attack on plain RSL was given in [21] when this problem was introduced. A more efficient attack was proposed in [18]. In particular, it showed that RSL can be solved in polynomial time for a number N of syndromes which is in general much smaller than the former bound $N \geq nr$ from [21].

Complexity of the [18] attack on plain RSL, where $a = \lfloor \frac{N}{r} \rfloor$

$$\begin{cases} \text{polynomial when } a - N/m \geq k, \text{ hence a fortiori when } N \geq (k+1)\frac{m}{m-r} \\ \mathcal{O}\left(q^{r(m - \lfloor \frac{m(n-k)-N}{n-a} \rfloor)}\right) \text{ otherwise.} \end{cases}$$

This attack exploits the fact that there exists an $\mathbb{F}_q$ -linear combination of the errors $\mathbf{e}_i$, $i \in \{1..N\}$ with $a = \lfloor \frac{N}{r} \rfloor$ zeroes in the leftmost positions. For instance, the goal is to find scalars $(\lambda_1, \dots, \lambda_\ell) \in \mathbb{F}_q^\ell$ and $\tilde{\mathbf{e}} \in E^{n-a}$ such that

$$(\mathbf{0} | \tilde{\mathbf{e}}) = \sum_{i=1}^{\ell} \lambda_i \mathbf{e}_i.$$

Then, the linear equation

$$(\mathbf{0} | \tilde{\mathbf{e}}) \mathbf{H}^\top = \sum_{i=1}^{\ell} \lambda_i \mathbf{s}_i$$

is rewritten as a linear system over $\mathbb{F}_q$ in $m(n-k)$ equations and $(n-a)m + N$ unknowns. When it is overdefined, solving this system takes polynomial time. Otherwise, [18] applies the same techniques as in combinatorial attacks on RD by sampling a random subspace F of dimension t . However, contrary to AGHT, the guess is successful when $E \subset F$ but not when $\alpha E \subset F$ for an arbitrary $\alpha \in \mathbb{F}_{q^m}^*$ (as we only consider $\mathbb{F}_q$ -linear combinations of the $\mathbf{e}_i$'s).

Adaptation to ℓ -RSL . We modify this algorithm in the same way as what we did for ℓ -RD. In the following, we restrict ourselves to the case when $n_1 = \dots = n_\ell = n$, $k = n$, $r = r_1 = \dots = r_\ell$ and $N \leq nr_1$. The condition on N implies that we cannot hope to “kill” completely one of the ℓ blocks of the error by putting zeroes. The complexity of this adaption is given below.

Complexity of our adaption on ℓ -RSL, where $a = \lfloor \frac{N}{r_1} \rfloor$
(when $n_1 = \dots = n_\ell = n$, $k = n$, $r = r_1 = \dots = r_\ell$ and $N \leq nr_1$)

$$\mathcal{O}\left(q^{r(m - \lfloor \frac{m(n\ell - \ell) - N - (\ell-1)nr}{n-a} \rfloor) + (\ell-1)r(m-r)}\right).$$

Proof. The condition on N makes that we cannot attack a support which is smaller than the common support E . Thus, we only care about fixing the maximum number of zeroes. Without loss of generality, we fix $a = \lfloor \frac{N}{r_1} \rfloor$ zeroes all in the first block. By doing so, the error $(\mathbf{0} | \tilde{\mathbf{e}})$ we end up with is still blockwise and

of the same support. We use the blockwise structure as in the AGHT adaptation. The probability of a correct guess $E_i \subset F_i$ for $i \in \{1..\ell\}$ is now

$$\prod_{i=1}^{\ell} q^{-r_i(m-t_i)},$$

and we want

$$(n-a)t_1 + \sum_{i=2}^{\ell} nt_i \leq m(n\ell - \ell) - N. \quad (4)$$

As the goal is to maximize the sum $\sum_{i=1}^{\ell} r_i t_i$ to maximize the probability that $E_i \subset F_i$ for $i \in \{1..\ell\}$, we take $t_i = r$ for $i > 1$, and thus $t_1 = \left\lfloor \frac{m(n\ell - \ell) - N - (\ell-1)nr}{n-a} \right\rfloor$, the highest value satisfying Equation 4. $\square$

5.3 A structural attack against 2-LRPC codes

It is also possible to consider structural attacks, by exploiting a possible particular structure of the code to recover the secret key $\mathbf{H}$. For example: in the case of an 2-LRPC code.

Proposition 5. *The complexity of recovering the structure of a 2-LRPC code $\mathcal{C}$ of size (n, n) by the Shortening and Truncating attack is estimated as:*

$$n^2 \cdot \min_{\substack{1 \leq t_1 \leq n \\ 1 \leq t_2 \leq n \\ t_1 + \lfloor n/d_1 \rfloor \leq u_1 \leq n \\ t_2 + \lfloor n/d_2 \rfloor \leq u_2 \leq n}} \left(q^{r_1 t_1} \times \mathcal{T}_{2\text{-RD}}((n - u_1, n), n - t_1 - \lfloor \frac{n}{d_1} \rfloor, (r_1, r_2), m), \right. \\ \left. q^{r_2 t_2} \times \mathcal{T}_{2\text{-RD}}((n, n - u_2), n - t_2 - \lfloor \frac{n}{d_2} \rfloor, (r_1, r_2), m) \right) \quad (5)$$

Proof. We explain using the attack described in [23] why we can reduce it to a subcode of $\mathcal{C}$ with smaller parameters.

Let $\mathbf{H} \in \mathbb{F}_{q^m}^{n \times 2n}$ the parity check matrix of $\mathcal{C}$. We can define the matrix as $\mathbf{H} = (\mathbf{H}_1 \mathbf{H}_2)$, where $\mathbf{H}_1, \mathbf{H}_2 \in \mathbb{F}_{q^m}^{n \times n}$ and $\mathbf{H}_1$ (resp. $\mathbf{H}_2$) has its coefficients belong to the same subspace F_1 (resp. F_2 , disjoint to F_1) of dimension d_1 (resp. d_2).

Let $\mathcal{D}$ the dual code of $\mathcal{C}$, whose $\mathbf{H} = (\mathbf{H}_1 \mathbf{H}_2)$ is a generator matrix. We denote by $(H_i)_{i \in \{1, \dots, n\}}$ the rows of $\mathbf{H}$, and we consider a word $\mathbf{x} \in \mathcal{D}$ obtained from linear combination in $\mathbb{F}_q$: $\mathbf{x} = \sum_{i=1}^n a_i H_i$, with $a_i \in \mathbb{F}_q$. Consider the block $\mathbf{H}_2$, whose coefficients belong to F_2 . Since F_2 has dimension d_2 , choose d_2 variables a_i correctly allows to put to 0 a coordinate of $\mathbf{x}$. Since there are n variables a_i , one can put to 0 with a good probability $\lfloor n/d_2 \rfloor$ coefficients of $\mathbf{x}$. Therefore, the dual code $\mathcal{C}^\perp$ contains with a good probability a word $\mathbf{x} = (\mathbf{x}_1 \mathbf{x}_2)$, whose

the coefficients of $\mathbf{x}_1$ belongs to F_1 and the $\lfloor n/d_2 \rfloor$ first coordinates of $\mathbf{x}_2$ are equal to zero (without loss of generality). Then, the attacker can perform the Shortening and Truncating attack on $\mathcal{D}$, knowing that the dimension of the code has already been reduced. $\square$

6 Algebraic attacks

The algebraic attacks of [31] on ℓ -RD are an adaptation of the known techniques for RD [14–16] by taking advantage of the block structure. They do not exploit the fact that the supports are pairwise disjoint. Since we introduce the ℓ -RSL problem, we also adapt the algebraic attack of [13] in a similar way. In this section, we will heavily rely on the fact that for a vector $\mathbf{x} \in \mathbb{F}_{q^m}^n$ and a basis $\beta \in \mathbb{F}_{q^m}$ for the extension field, there exists a unique matrix $\mathbf{M}(\mathbf{x}) \in \mathbb{F}_q^{m \times n}$ such that $\mathbf{x} = \beta \mathbf{M}(\mathbf{x})$.

6.1 MaxMinors attack

As in the most recent combinatorial attacks, RD is reduced to the problem of finding a weight r codeword in the code $\mathcal{C}_{\mathbf{y}} \stackrel{\text{def}}{=} \mathcal{C} \oplus \langle \mathbf{y} \rangle_{\mathbb{F}_{q^m}}$. The error vector satisfies the equation

$$\mathbf{e} \mathbf{H}_{\mathbf{y}}^T = \mathbf{0},$$

where $\mathbf{H}_{\mathbf{y}} \in \mathbb{F}_{q^m}^{(n-k-1) \times n}$ is a systematic parity-check matrix for $\mathcal{C}_{\mathbf{y}}$. We then express $\mathbf{M}(\mathbf{e}) \in \mathbb{F}_q^{m \times n}$ as a product $\mathbf{S} \mathbf{C}$, where $\mathbf{S} \in \mathbb{F}_q^{m \times r}$ and $\mathbf{C} \in \mathbb{F}_q^{r \times n}$ are the support and coefficient matrices respectively. Finally, the matrix $\mathbf{S} \mathbf{C} \mathbf{H}_{\mathbf{y}}^T \in \mathbb{F}_{q^m}^{r \times (n-k-1)}$ is not full-rank because $\beta \mathbf{S} \mathbf{C} \mathbf{H}_{\mathbf{y}}^T = \mathbf{0}$.

Modeling 1 (MaxMinors) *Let $\mathbf{H}_{\mathbf{y}} \in \mathbb{F}_{q^m}^{(n-k-1) \times n}$ be a systematic parity-check matrix for $\mathcal{C}_{\mathbf{y}} = \mathcal{C} \oplus \langle \mathbf{y} \rangle_{\mathbb{F}_{q^m}}$ and let $\mathbf{C} \in \mathbb{F}_q^{r \times n}$ be the secret coefficient matrix associated to $\mathbf{e} \in \mathbb{F}_{q^m}^n$. The MaxMinors modeling is the system defined by $\{P_J\}_{J \subset \{1..n-k-1\}, \#J=r}$, where*

$$P_J \stackrel{\text{def}}{=} \left| \mathbf{C}(\mathbf{H}_{\mathbf{y}}^T)_{*,J} \right|.$$

By using the Cauchy-Binet formula, this system is known to be linear (over $\mathbb{F}_{q^m}$) in the maximal minors $c_T \stackrel{\text{def}}{=} |\mathbf{C}|_{*,T}$ of $\mathbf{C}$ for $T \subset \{1..n\}$, $\#T = r$. As these minors are over $\mathbb{F}_q$, the attack proceeds by solving a system projected over $\mathbb{F}_q$ containing $m \binom{n-k-1}{r}$ equations.

In order to solve ℓ -RD, [31] propose to fix certain variables in the MaxMinors system. A previous attempt of the same type can be found in the RQC submission on non-homogeneous errors [1]. To attack an ℓ -RD instance of block size

$n \stackrel{\text{def}}{=} \sum_{i=1}^{\ell} n_i$ and dimension k with $r \stackrel{\text{def}}{=} \sum_{i=1}^{\ell} r_i$, the idea is to write the coefficient matrix as

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_1 & & & \\ & \mathbf{C}_2 & & \\ & & \ddots & \\ & & & \mathbf{C}_{\ell} \end{pmatrix} \in \mathbb{F}_q^{r \times n}, \quad \mathbf{C}_i \in \mathbb{F}_q^{r_i \times n_i}. \quad (6)$$

If we set $n_{\leq j} \stackrel{\text{def}}{=} \sum_{i=1}^j n_i$, we notice that the minor variables that are possibly non-zero are such that $T_j \stackrel{\text{def}}{=} (T - n_{\leq j-1}) \cap \{1..n_j\}$ is of size r_j for $j \in \{1..\ell\}$. This allows to consider $\prod_{i=1}^{\ell} \binom{n_i}{r_i}$ unknowns instead of $\binom{n}{r}$. Moreover, such minors can be seen as product of smaller ones, i.e.,

$$c_T = \prod_{i=1}^{\ell} c_{i, T_i}, \quad c_{i, T_i} \stackrel{\text{def}}{=} |\mathbf{C}_i|_{*, T_i}. \quad (7)$$

The question left open in [31] is the study of linear dependencies between the MaxMinor equations by zeroing the rest of the variables. We attempted to study such relations in the system over $\mathbb{F}_{q^m}$ and a sketch of analysis is presented below.

As explained above, we restrict ourselves to the c_T variables as in Equation (7) such that the set T_i is of size r_i for $i \in \{1..\ell\}$. Such variables will be referred to as *admissible*. For the sake of simplicity and to stick to our choice of parameters, we will mostly consider the cases $\ell = 2$ and $\ell = 3$ with blocks of the same size, i.e., $n_i = n$ for any $i \in \{1..\ell\}$. We start by recalling the following result of [16] on the shape of the P_J equations that is valid for a code of parameters $[\ell n, n]_{\mathbb{F}_{q^m}}$ and an error of weight r , regardless of the error pattern.

Lemma 1 (Proposition 2 in [16]). *For $J \subset \{1..(\ell - 1)n - 1\}$, $\#J = r$, we have*

$$P_J = c_{J+n+1} + \sum_{\substack{T^- \subset \{1..n+1\}, T^+ \subset (J+n+1) \\ T = T^- \cup T^+, \#T=r, T^- \neq \emptyset}} c_T |\mathbf{H}_{\mathbf{y}}|_{J, T}. \quad (8)$$

We may adopt the same variable order as in [16]. It is defined by $c_T \prec c_{T'}$ for ordered subsets $T = \{t_1 < \dots < t_r\}$ and $T' = \{t'_1 < \dots < t'_r\}$ if $t_i = t'_i$ up to some index i_0 where $t'_{i_0} > t_{i_0}$. In this case, the variable c_{J+n+1} corresponds to the leading monomial of P_J . We will call *a-admissible* the admissible variables c_T such that $n+1 \notin T$ and *b-admissible* those such that $n+1 \in T$. For an *a*-admissible variable c_T to appear in P_J , Lemma 1 shows that we must have $T_i \subset J+n+1$ for $i \geq 2$ but there is no constraint on T_1 . Similarly, for a *b*-admissible variable c_T to appear in the same equation we must have $T_2 \setminus \{n+1\} \subset J+n+1$ and $T_i \subset J+n+1$ for $i \in \{3..\ell\}$. If we set $J_2 \stackrel{\text{def}}{=} J \cap \{1..n-1\}$ and $J_i \stackrel{\text{def}}{=} J \cap \{(i-2)n..(i-1)n-1\}$ for $i \in \{3..\ell\}$, we can rewrite these conditions as $T_2 - (n+1) \subset J_2$ and $T_i - (n+1) \subset J_i$ for $i \geq 3$ for an *a*-admissible variable c_T

and $T_2 \setminus \{n+1\} - (n+1) \subset J_2$ and $T_i - (n+1) \subset J_i$ for $i \geq 3$ for a b -admissible variable c_T . This implies that the monomial content of P_J after zeroing highly depends on the size of the J_i 's. More precisely, the above discussion leads to

Lemma 2. *For any subset $J \subset \{1..(\ell-1)n-1\}$ of size $r = \sum_{i=1}^{\ell} r_i$, let $J_2 \stackrel{\text{def}}{=} J \cap \{1..n-1\}$ and let $J_i \stackrel{\text{def}}{=} J \cap \{(i-2)n..(i-1)n-1\}$ for $i \in \{3..\ell\}$.*

- *If $\#J_2 \leq r_2 - 2$ or if $\#J_i \leq r_i - 1$ for some index $i \in \{3..\ell\}$, the P_J equation becomes zero after setting the non-admissible variables to zero.*
- *If $\#J_2 = r_2 - 1$ and if $\#J_i \geq r_i$ for all $i \in \{3..\ell\}$, the P_J equation only contains b -admissible variables after this operation.*
- *If $\#J_2 \geq r_2$ and if $\#J_i \geq r_i$ for all $i \in \{3..\ell\}$, the same equation contains both a -admissible and b -admissible variables.*

When $\ell = 2$, the first case in Lemma 2 never occurs so there is no trivial loss in the number of equations available after zeroing. However, the former leading variable c_{J+n+1} in P_J is always zeroed and several equations now have the same leading monomial. For instance, as long as the associated minor of $\mathbf{H}_y$ is non-zero, the variable $c_{1,\{n_1-r_1+1..n_1\}c_{2,T_2}}$ will be the greatest one in any P_J equation such that the greatest r_2 elements of the set $J+n+1$ correspond to T_2 . In spite of this, we observed in our tests that all the equations remained linearly independent, see Table 1 (we tested both systems over $\mathbb{F}_{q^m}$ and over $\mathbb{F}_q$). As there are m times more equations, the experiments over $\mathbb{F}_q$ were more difficult to conduct due to the memory limit.

m	n	r_1	r_2	MaxMinors $\mathbb{F}_{q^m}$	MaxMinors $\mathbb{F}_q$
10	10	2	2	LI	LI
11	11	2	2	LI	LI
12	12	2	2	LI	segfault
11	11	3	2	LI	segfault
14	14	3	2	LI	segfault
16	16	3	2	LI	segfault

Table 1: Experiments on MaxMinors systems over $\mathbb{F}_{q^m}$ and over $\mathbb{F}_q$ for a code of parameters $[2n, n]_{\mathbb{F}_{q^m}}$ and blocks of size n . We write “LI” when the equations appeared to be linearly independent.

When $\ell = 3$, the number of linearly independent equations over $\mathbb{F}_{q^m}$ is decreased because there are subsets J of size $r_1 + r_2 + r_3$ such that $\#J_2 \leq r_2 - 2$ or $\#J_3 \leq r_3 - 1$. Lemma 2 in fact shows that it is bounded from above by

$$V_3(n, r_1, r_2, r_3) \stackrel{\text{def}}{=} \sum_{j=r_2-1}^{r_1+r_2} \binom{n-1}{j} \binom{n}{r_1+r_2+r_3-j}.$$

From our experiments, it also seems bounded from below by

$$L_3(n, r_1, r_2, r_3) \stackrel{def}{=} \sum_{j=r_2}^{r_1+r_2} \binom{n-1}{j} \binom{n}{r_1+r_2+r_3-j}.$$

We can be a bit more precise. Among the $\binom{n-1}{r_2-1} \binom{n}{r_1+r_3+1}$ equations that only contain b -admissible monomials, it seems that $\binom{n-1}{r_2-1} \binom{n-1}{r_1+r_3}$ of them are linearly independent. Furthermore, on their own, the rest of the P_J equations such that $\#J_2 \geq r_2$ seem to be linearly independent. We have $\sum_{j=r_2}^{r_1+r_2} \binom{n-1}{j} \binom{n}{r_1+r_2+r_3-j}$ of them, which gives an upper bound

$$U_3(n, r_1, r_2, r_3) \stackrel{def}{=} \sum_{j=r_2}^{r_1+r_2} \binom{n-1}{j} \binom{n}{r_1+r_2+r_3-j} + \binom{n-1}{r_2-1} \binom{n-1}{r_1+r_3} \quad (9)$$

which is tighter than $V_3(n, r_1, r_2, r_3)$. However, it is still not tight because there are extra linear relations when we combine both groups of equations.

n	r_1	r_2	r_3	L_3	expe	U_3
6	2	2	2	425	444	450
7	2	2	2	1540	1587	1622
8	2	2	2	4410	4480	4600
9	2	2	2	10752	10780	11760*

Table 2: Experimental number of linearly independent equations in the MaxMinors system over $\mathbb{F}_{q^m}$ for a code of parameters $[3n, n]_{\mathbb{F}_{q^m}}$ and blocks of size n . We do not give the value of m since it does not have any influence on this number.

We now move on to our estimates in the plain and in the hybrid setting. We have not analyzed the hybrid setting in details. Thus, we assume that the specialization of [16, §4.3] does not induce extra linear relations (this is in accordance with [15, §5] in the random case and with [18, Corollary 1] in the RQC case).

Message attack. Estimate 1 is based upon the assumption that the equations remain linearly independent when $\ell = 2$, which is what we observed in our tests.

We set $N_2(n, r_1, r_2) \stackrel{def}{=} \binom{n-1}{r_1+r_2}$.

Estimate 1 (2 blocks) *We expect to solve a 2-RD instance of parameters $(m, n_1 = n, n_2 = n, k = n, (r_1, r_2))$ by Gaussian elimination on the MaxMinors system whenever*

$$mN_2(n, r_1, r_2) \geq \binom{n}{r_1} \binom{n}{r_2} - 1, \quad (10)$$

with cost $\mathcal{O}\left(mN_2(n, r_1, r_2) \binom{n}{r_1}^{\omega-1} \binom{n}{r_2}^{\omega-1}\right)$, $2 \leq \omega \leq 3$. When Equation (10) does not hold, we estimate the cost of the hybrid approach of by

$$\mathcal{O}\left(\min_{\substack{(a_1, a_2) \\ mN_2(n, r_1, r_2) \geq \binom{n-a_1}{r_1} \binom{n-a_2}{r_2} - 1}} \left(q^{a_1 r_1 + a_2 r_2} mN_2(n, r_1, r_2) \binom{n-a_1}{r_1}^{\omega-1} \binom{n-a_2}{r_2}^{\omega-1}\right)\right).$$

When $\ell = 3$, we take $mN_3(n, r_1, r_2, r_3) \stackrel{def}{=} mU_3(n, r_1, r_2, r_3)$, where the value $U_3(n, r_1, r_2, r_3)$ defined in Equation (9) is a non-tight upper bound on the number of linearly independent equations over $\mathbb{F}_{q^m}$ and where the m factor assumes that the equations projected over $\mathbb{F}_q$ remain linearly independent. On our parameters, this value is still quite close to the maximum number of equations $m \binom{2n-1}{r_1+r_2}$.

Estimate 2 (3 blocks) We expect to solve a 3-RD instance of parameters $(m, n_1 = n, n_2 = n, n_3 = n, k = n, (r_1, r_2, r_3))$ by Gaussian elimination on the MaxMinors system whenever

$$mN_3(n, r_1, r_2, r_3) \geq \binom{n}{r_1} \binom{n}{r_2} \binom{n}{r_3} - 1, \quad (11)$$

with cost $\mathcal{O}\left(mN_3(n, r_1, r_2, r_3) \binom{n}{r_1}^{\omega-1} \binom{n}{r_2}^{\omega-1} \binom{n}{r_3}^{\omega-1}\right)$, $2 \leq \omega \leq 3$. When Equation (11) does not hold, we estimate the cost of the hybrid approach of by

$$\mathcal{O}\left(\min_{\substack{(a_1, a_2, a_3) \\ mN_3(n, r_1, r_2, r_3) \geq \binom{n-a_1}{r_1} \binom{n-a_2}{r_2} \binom{n-a_3}{r_3} - 1}} \left(q^{a_1 r_1 + a_2 r_2 + a_3 r_3} mN_3(n, r_1, r_2, r_3) \binom{n-a_1}{r_1}^{\omega-1} \binom{n-a_2}{r_2}^{\omega-1} \binom{n-a_3}{r_3}^{\omega-1}\right)\right)$$

Structural attack. In this case, we have more freedom to fix coordinates to zero in the error vector. We reduce to a problem with a unique solution with probability 1 and we then proceed as before. On an instance with parameters $(m, n_1 = n, n_2 = n, k = n, (d_1, d_2))$, we can freely

- fix b_1 on the left and then the rest $b_2 = \left\lfloor \frac{n_1 + n_2 - k - r_1 b_1}{r_2} \right\rfloor$ on the right;
- fix b_2 zeroes on the right first and then $b_1 = \left\lfloor \frac{n_1 + n_2 - k - r_2 b_2}{r_1} \right\rfloor$ on the left.

By doing so, we expect to attack a new instance with block size $n_1 = n - b_1$, $n_2 = n - b_2$ and with dimension $n - b_1 - b_2$. The codimension remains $(2n - b_1 - b_2) - (n - b_1 - b_2) = n$.

Estimate 3 The complexity of this attack is $\mathcal{O}(m \times \min(A, B))$, where

$$A = \min_{\substack{0 \leq b_1 \leq \lfloor n/d_1 \rfloor \\ b_2 = \left\lfloor \frac{n - r_1 b_1}{d_2} \right\rfloor}} \left(\min_{mN_2(n, d_1, d_2) \geq \binom{n-b_1-a_1}{d_1} \binom{n-b_2-a_2}{d_2} - 1} q^{a_1 d_1 + a_2 d_2} N_2(n, d_1, d_2) \binom{n-b_1-a_1}{d_1}^{\omega-1} \binom{n-b_2-a_2}{d_2}^{\omega-1} \right)$$

$$B = \min_{\substack{0 \leq b_2 \leq \lfloor n/d_2 \rfloor \\ b_1 = \left\lfloor \frac{n - d_2 b_2}{d_1} \right\rfloor}} \left(\min_{mN_2(n, d_1, d_2) \geq \binom{n-b_1-a_1}{d_1} \binom{n-b_2-a_2}{d_2} - 1} q^{a_1 d_1 + a_2 d_2} N_2(n, d_1, d_2) \binom{n-b_1-a_1}{d_1}^{\omega-1} \binom{n-b_2-a_2}{d_2}^{\omega-1} \right)$$

6.2 Attack based on Support-Minors

The Support-Minors system was introduced in [16] as a new modeling for the MinRank problem but its analysis in the context of RD was inaccurate. This was corrected in [15] where they propose the SM- $\mathbb{F}_{q^m}^+$ attack. When MaxMinors projected over $\mathbb{F}_q$ cannot be solved by direct linearization, it consists in adding the following equations:

Modeling 2 (Support-Minors for RD) Let $G \in \mathbb{F}_{q^m}^{k \times n}$ be a systematic generator matrix of C and let $C \in \mathbb{F}_q^{r \times n}$ be the secret coefficient matrix associated to $e \in \mathbb{F}_{q^m}^n$. The Support-Minors modeling is the system containing the equations $\{Q_I\}_{I \subset \{1..n\}, \#I=r+1}$, where

$$Q_I \stackrel{\text{def}}{=} \left| \begin{pmatrix} xG + y \\ C \end{pmatrix}_{*,I} \right|.$$

This is a bilinear system in $c_T \in \mathbb{F}_q$ and $x_j \in \mathbb{F}_{q^m}$ for $j \in \{1..k\}$.

On some RD instances, it can lead to better complexities than the hybrid MaxMinors attack. In the random case, it is shown in [15] that some Support-Minors equations are redundant with the MaxMinors modeling and that we can restrict ourselves to the subsets I such that $I \cap \{1..k+1\}$ is of size ≥ 2 . The Q_I equations corresponding to these subsets are linearly independent and [15, Proposition 2] states that the leading monomial of Q_I with respect to $\prec$ is equal to $x_{\min(I)} c_{I \setminus \{\min(I)\}}$ (the quantity $x_{\min(I)}$ is well defined since $\min(I) \leq k$ for such subsets).

However, we observe that Support-Minors is much sparser than MaxMinors. In particular, a lot more relations are to be expected when we apply it to ℓ -RD. By Laplace expansion along the first row, the c_T variables present in Q_I are included in the set $\{c_{I \setminus \{i\}}, i \in I\}$. Now, a $c_{I \setminus \{i\}}$ that remains after specialization is necessarily as in Equation (7). In other words, this means that $(I \setminus \{i\} - n_{\leq j-1}) \cap \{1..n_j\}$ is of size r_j for all j . It imposes that $(I - n_{\leq j-1}) \cap \{1..n_j\}$ is of size r_j except for one j where it is of size $r_j + 1$. Conversely, for such an I and j_0 for which $(I - n_{\leq j_0-1}) \cap \{1..n_{j_0}\}$ is of size $r_{j_0} + 1$ and the rest of the intersections are of size r_j , the c_T present are of the form $c_{I \setminus \{i\}}, i \in I \cap \{n_{\leq j_0-1} + 1..n_{\leq j_0}\}$.

For $\ell = 2$ and $\ell = 3$ with blocks of the same size n , the number of linearly independent equations is bounded from above by $\binom{n}{r_1+1} \binom{n}{r_2} + \binom{n}{r_1} \binom{n}{r_2+1}$ and $\binom{n}{r_1+1} \binom{n}{r_2} \binom{n}{r_3} + \binom{n}{r_1} \binom{n}{r_2+1} \binom{n}{r_3} + \binom{n}{r_1} \binom{n}{r_2} \binom{n}{r_3+1}$ instead of $\binom{2n}{r_1+r_2+1}$ and $\binom{3n}{r_1+r_2+r_3+1}$ respectively. Furthermore, for a code of parameters $[\ell n, n]_{\mathbb{F}_{q^m}}$, the Q_I equations that keep the same leading term after restriction to the admissible c_T variables correspond to the subsets I such that $\#(I \cap \{1..n\}) = r_1 + 1$. Otherwise, the largest admissible c_T variable present in Q_I is $c_{I \setminus \{j\}}$, where j is the minimum of the unique set $I \cap \{(i-1)n + 1..in\}, i \in \{1..\ell\}$ which is of size $r_i + 1$ (this is indeed $c_{I \setminus \{\min(I)\}}$ if $\#(I \cap \{1..n\}) = r_1 + 1$). However, deriving the precise number of linearly independent equations seems more difficult and a fortiori the rank of the full SM- $\mathbb{F}_{q^m}^+$ modeling.

For this reason and as the progress over MaxMinors in the random case was often only by a few bits, we adopt Estimate 4:

Estimate 4 *We do not take into account $SM\text{-}\mathbb{F}_{q^m}^+$ to derive our parameters.*

6.3 Algebraic attack on ℓ -RSL

We start by describing the approach of [13] on a plain RSL instance. As in the above combinatorial attack, it targets a specific vector $\mathbf{e} \in \mathbb{F}_{q^m}^n$ which is a linear combination over $\mathbb{F}_q$ between the N errors $\mathbf{e}^{(i)}$, $i \in \{1..N\}$. By keeping the same notation as in the RD case, we may write

$$\mathbf{e}\mathbf{H}^\top = \left(\sum_{i=1}^N \lambda_i \mathbf{e}^{(i)} \right) \mathbf{H}^\top = \left(\sum_{i=1}^N \lambda_i \beta \mathbf{S} \mathbf{C}^{(i)} \right) \mathbf{H}^\top = \beta \mathbf{S} \mathbf{C} \mathbf{H}^\top, \quad (12)$$

where $\mathbf{S} \in \mathbb{F}_q^{m \times r}$ is the support matrix common to all the errors, where $\mathbf{C}^{(i)} \in \mathbb{F}_q^{r \times n}$ is the coefficient matrix of $\mathbf{e}_i$ and where $\mathbf{C} \stackrel{\text{def}}{=} \sum_{i=1}^N \lambda_i \mathbf{C}^{(i)}$. In order to solve a problem with a unique solution, [13] targets a vector $\mathbf{e}$ such that the matrix $\mathbf{C}$ is of rank $< r$ and/or contains zero columns (corresponding to zeroes in $\mathbf{e}$). To be consistent with what was presented in the combinatorial attack, we will restrict ourselves to looking for a full-rank matrix $\mathbf{C}$ which contains as many zero columns as possible to belong to the space generated by the $\mathbf{C}_i$'s, i.e., $a = \lfloor \frac{N}{r} \rfloor$. In other words, we will consider $\mathbf{C} \stackrel{\text{def}}{=} \begin{pmatrix} \mathbf{0}_{a \times r} & \tilde{\mathbf{C}} \end{pmatrix}$, where $\tilde{\mathbf{C}} \in \mathbb{F}_q^{r \times (n-a)}$ is of full-rank. Note that $\mathbf{C}\mathbf{H}^\top = \tilde{\mathbf{C}}\tilde{\mathbf{H}}^\top$, where $\tilde{\mathbf{H}} \stackrel{\text{def}}{=} \mathbf{H}_{*,[a+1,n]}$. For $i \in \{1..N\}$, let $\mathbf{s}^{(i)} \in \mathbb{F}_{q^m}^{n-k}$ be the syndrome associated to $\mathbf{e}^{(i)}$. By Equation (12), the syndrome $\mathbf{e}\mathbf{H}^\top = \sum_{i=1}^N \lambda_i \mathbf{s}^{(i)}$ is a linear combination over $\mathbb{F}_{q^m}$ between the rows of $\mathbf{C}\mathbf{H}^\top$. Thus, the matrix

$$\Delta \stackrel{\text{def}}{=} \begin{pmatrix} \sum_{i=1}^N \lambda_i \mathbf{s}^{(i)} \\ \tilde{\mathbf{C}}\tilde{\mathbf{H}}^\top \end{pmatrix} \in \mathbb{F}_{q^m}^{(r+1) \times (n-k)}$$

is of rank at most r .

Modeling 3 (RSL-Minors) *Let $a = \lfloor \frac{N}{r} \rfloor$, let $\tilde{\mathbf{C}} \in \mathbb{F}_q^{r \times (n-a)}$ be the coefficient matrix associated to the secret $\tilde{\mathbf{e}}$ in the target vector $\mathbf{e} = (\mathbf{0} | \tilde{\mathbf{e}})$ and let $\tilde{\mathbf{H}} \stackrel{\text{def}}{=} \mathbf{H}_{*,[a+1,n]}$. The RSL-Minors modeling is defined by $\{\Delta_J\}_{J \subset \{1..n-k\}, \#J=r+1}$, where*

$$\Delta_J \stackrel{\text{def}}{=} |\Delta_{*,J}| = \left| \begin{pmatrix} \sum_{i=1}^N \lambda_i \mathbf{s}^{(i)} \\ \tilde{\mathbf{C}}\tilde{\mathbf{H}}^\top \end{pmatrix}_{*,J} \right|.$$

Using the Cauchy-Binet formula, this system can be seen as bilinear in the λ_i variables and the maximal minors of $\tilde{\mathbf{C}}$ (that we still denote by c_T).

Once again, as the equations have coefficients in $\mathbb{F}_{q^m}$ and as the variables are searched in $\mathbb{F}_q$, [13] solves a system projected over $\mathbb{F}_q$ containing $m \binom{n-k}{r+1}$ equations.

In the ℓ -RSL case, all the coefficient matrices $\mathbf{C}^{(i)}$ are block diagonal as in Equation (6). This property is preserved by linear combination, which means that we can use the same specialization as in the ℓ -RD case. The adaptation of the above would then be to target a matrix $\mathbf{C}$ such that the j -th diagonal block $\mathbf{C}_j$ contains a_j zero columns, for $j \in \{1..\ell\}$, under the constraint $\sum_{j=1}^{\ell} a_j r_j \leq N$. Assuming that the number of linearly independent equations remains the same in all cases, we would like to minimize the number of non-zero c_T variables $\prod_{i=1}^{\ell} \binom{n_i - a_i}{r_i}$. Note that there is no formula for this minimum in the general case and that some particular ways of fixing zero columns might create algebraic relations.

For the parameter we consider (see the Section 8), the number of given syndromes is very low, and far from being big enough, so that the attacks based on the ℓ -RSL problem impacts the security. In practice, for the parameters we consider, the best attacks are the attacks against ℓ -RSD problem.

7 Application to cryptanalysis

In this section, we apply the above attacks on the parameters given by [31] for their improvement of Lake (ROLLO-I), based on 2-LRPC codes. There are two types of attacks to consider for the security of their parameters, the structural attacks targeting weights (d_1, d_2) and the message attacks targeting weights (r_1, r_2) . In our case we propose two new structural attacks to recover the secret key of the system.

A first attack (attack1) corresponds to the attack against 2-LRPC codes explained in Section 5.3. The idea of the attack is to shorten as much as possible the block corresponding to the higher d_i , then shorten on these $\frac{n}{d_i}$ positions and then truncate the block corresponding to d_i , then one gets an homogeneous error that we can attack with algebraic attacks for homogeneous errors. It is also possible to increase the number of terms shortened by guessing zero positions on the d_i part at a cost of 2^{d_i} per new zero coordinate. In practice the best results are obtained when guessing sufficiently many more zeros coordinates the part corresponding to the case where the MaxMinor attack is the most efficient, in that case we estimated the polynomial part at the cost of n^2 as it is usually the case for attacks and parameters and also we consider $w = 2.8$ the Strassen exponent.

A second attack consists in having the same Shortening and Truncating approach but rather than truncating, we just attack directly the code with algebraic attacks for blockwise errors described in [31], notice that at the difference of Attack1, it is more efficient to shorten on the smallest r_i , which permits to better decrease the dimension of the code.

The table in Figure 7 gathers the complexities of our cryptanalyses of parameters on Lake, given by [31], and their claimed security.

n	m	(d_1, d_2)	(r_1, r_2)	Security	Claimed M.A.S.	Claimed S.A.S.	Attack 1	Attack 2
67	61	(5,4)	(4,4)	128	145	160	132	116
79	71	(5,5)	(5,5)	192	225	255	181	166
89	79	(6,5)	(5,5)	256	281	266	246	224

Fig. 7: Security of parameters on Lake given by [31]. We refers as M.A.S. (resp. S.A.S.) for Message (resp. Structural) Attack Security.

Our new attack is very efficient against LAKE parameters given in [31], outperforming by 44 bits the security for structural attacks for the 128 bits NIST type parameters.

8 Parameters

We discuss here on the security and parameters of our two new schemes. For all our protocols, the parameters we propose are compliant with NIST security levels 1 and 3 of 143 and 207 classical bit security. Two sets of parameters are proposed for each of the schemes: the first designed to resist attacks with $\omega = 2.8$ as the Strassen constant (value with which common attacks are considered), the second (still compliant with the NIST security definition of Level 1 and Level 3) corresponds to a higher security constraint with $\omega = 2$, for which no practical attack is known for the moment.

To have available both several syndromes and blockwise errors allows to achieve excellent sizes: the first idea allows to obtain more coordinates to guess the support error, and the second gives syndromes relying to smaller spaces, which makes decoding easier.

8.1 Parameters of ILRPC-Block-MS

The security of the scheme relies on the hardness to solve the instance of a 2-IRSL problem on a code $[2n_2, n_2]_{q^m}$ with parity check matrix: $(\mathbf{1} \ \mathbf{h})$, where n_1 syndromes with the same block support of size (n_2, n_2) and dimension (r_1, r_2) are given in input. However, the attacks against 2-IRSL are not the best because the number of syndromes given is too small within the parameters we propose. One refers to this attack as Attack 1. One must also consider the structural attack against LRPC (Attack 2).

Parameters and resulting sizes are presented in Figure 8 for $\omega = 2.8$, and in Figure 9 for $\omega = 2$. Since the ideal parity check matrix is completely determined

by the polynomial $\mathbf{h}$, its size is reduced to $\lfloor \frac{n_2 m}{8} \rfloor$ bytes. The $\mathbf{c}$ is made of n_1 polynomials of degree n_2 whose coefficients belong to $\mathbb{F}_{q^m}$, so its size is $\lfloor \frac{n_1 n_2 m}{8} \rfloor$ bytes. The parameters we obtain compare very well with previous results: 3.8 kB for 128 bits security in [31] and 2.4 kB for the multiple syndromes approach [3]. Indeed as explained in the introductory section, the blockwise approach is essentially interesting for RQC and less for LRPC, since blockwise small weight errors are more vulnerable to the Shortening and Truncating approach of Section 5, indeed the smallest the d_i the greater the zeros set for shortening. Overall the approach becomes more interesting when one considers the XMS approach (originally described in [3]) that uses an extended decoding algorithm for LRPC, decoding algorithm that we generalize in Section 3 to the case of blockwise rank errors.

Scheme	m	n_2	(d_1, d_2)	(r_1, r_2)	n_1	DFR	Att. 1	Att. 2	pk + ct (kB)
ILRPC-Block-xMS-128 ($r + 3$)	59	84	(5,5)	(4,4)	2	-128	154	176	1.86
ILRPC-Block-xMS-128 ($r + 5$)	53	84	(5,5)	(4,4)	2	-128	162	185	1.67
ILRPC-Block-xMS-192 ($r + 2$)	83	83	(6,5)	(5,5)	3	-192	242	204	3.45
ILRPC-Block-xMS-192 ($r + 3$)	83	79	(6,5)	(5,5)	3	-194	235	202	3.28

Fig. 8: Comparaison of parameters of ILRPC schemes, security for $\omega = 2.8$

Scheme	m	n_2	(d_1, d_2)	(r_1, r_2)	n_1	DFR	Att. 1	Att. 2	pk + ct (kB)
ILRPC-Block-xMS-128 ($r + 4$)	61	95	(5,5)	(5,4)	2	-145	179	147	2.17
ILRPC-Block-xMS-128 ($r + 6$)	59	89	(5,5)	(5,4)	2	-133	177	145	1.97
ILRPC-Block-xMS-192 ($r + 2$)	89	84	(6,6)	(5,5)	3	-192	204	213	3.74
ILRPC-Block-xMS-192 ($r + 3$)	83	85	(6,6)	(5,5)	3	-195	209	213	3.53

Fig. 9: Parameters for ILRPC schemes with $\omega = 2$

8.2 Parameters of RQC-Block-MS-AG scheme

The attacks 1 and 2 relies on the algebraic attack which consists on solving the 2-IRSD (on the $[2n_2, n_2]_{q^m}$ ideal code with parity check matrix $(\mathbf{1} \ \mathbf{h})$) and 3-IRSL problem (on the $[3n_2, n_2]_{q^m}$ ideal code whose $\begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{h} \\ \mathbf{0} & \mathbf{1} & \mathbf{s} \end{pmatrix}$ is a parity check matrix). The attack 3 is the Shortening and Truncating attack on the 2-IRSD instance. Note that there is currently no attack that takes advantage of the ideal structure of the parity check matrix, this is why these instances are considered as difficult to solve as 2-RSD and 3-RSL instances.

The decoding algorithm takes as input n_2 vectors having the same errors support, that is to say it has $n_1 n_2$ available coordinates to compute the support. We use a

public Augmented Gabidulin code of length $n_1 n_2$ and dimension k , constructed from a vector $\mathbf{g}$ of size m . Let ε the number of erasure coordinates one uses to recover the support error. The values above must be chosen such that the decoding capacity of the code thus obtained: $\delta = \lfloor \frac{m-k+\varepsilon}{2} \rfloor$, must be greater than or equal to the weight of the error which is $r_{\mathbf{x}} r_1 + r_{\mathbf{y}} r_2 + r_{\mathbf{e}}$. On the other hand, the resulting decryption failure rate (see Proposition 1) must be remain low.

The resulting parameters for 128 and 192 bits of security are presented in Figure 10, and in Figure 11 for the optimized version in which 1 belongs to the support of $\mathbf{R}_2$. The sizes are computing according to the following formulas: $|\mathbf{pk}| = \lceil \frac{n_2 m}{8} \rceil + \frac{2\lambda}{8}$ and $|\mathbf{ct}| = \lceil \frac{2n_1 n_2 m}{8} \rceil$. Since $\mathbf{g}$ and $\mathbf{h}$ are uniformly sampled from their respective spaces, they can be represented as seeds of size λ bits. The ciphertext $\mathbf{ct}$ contains two matrices lying in $\mathbb{F}_{q^m}^{n_2 \times n_1}$. The decrease in size of public key and ciphertext over time is a direct consequence of the decrease in the size of the parameters.

Scheme	m	n_2	q	k	ε	r_1	r_2	$r_{\mathbf{x}}$	$r_{\mathbf{y}}$	$r_{\mathbf{e}}$	n_1	Att. 1	Att. 2	Att. 3	DFR	pk + ct (kB)
RQC-Block-MS-AG-128	43	52	2	3	32	4	4	4	4	4	2	145	153	154	-145	1.43
RQC-Block-MS-AG-192	67	70	2	3	45	5	5	5	5	6	2	232	207	234	-196	2.98

Fig. 10: Parameters for RQC-Block-MS-AG, $\omega = 2.8$

Scheme	m	n_2	q	k	ε	r_1	r_2	$r_{\mathbf{x}}$	$r_{\mathbf{y}}$	$r_{\mathbf{e}}$	n_1	Att. 1	Att. 2	Att. 3	DFR	pk + ct (kB)
RQC-Block-MS-AG-128	43	52	2	3	32	4	4	4	4	4	2	145	153	154	-168	1.43
RQC-Block-MS-AG-192	71	64	2	3	46	5	5	5	5	7	2	219	214	228	-202	2.89

Fig. 11: Parameters for RQC-Block-MS-AG, $\omega = 2.8$, $1 \in \text{Supp } \mathbf{R}_2$

Note that a limitation for 128 bits of security comes from the attack on the 2-IRSD instance, this is why the optimization gives no advantage. Conversely, it benefits to the parameters for 192 bits of security.

For fair comparison with the standard RQC protocol, we also provide parameters with an only syndrome allowing a better understanding of the benefit of taking block errors (see Figure 12 and Figure 13).

We also present the parameters of previous versions of RQC in Figure 14. We observe that the different developments have made it possible to consider increasingly smaller parameters, particularly due to the weight of the error in the message to decode which decreases for the same security.

Scheme	m	n_2	q	k	ε	r_1	r_2	r_x	r_y	r_e	Att. 1	Att. 2	DFR	pk + ct (kB)
RQC-Block-AG-128	47	89	2	3	28	4	4	4	4	4	172	178	-133	1.60
RQC-Block-AG-192	61	128	2	4	43	4	5	5	5	5	247	235	-198	2.98

Fig. 12: Parameters for RQC-Block-AG, $\omega = 2.8$

Scheme	m	n_2	q	k	ε	r_1	r_2	r_x	r_y	r_e	Att. 1	Att. 2	DFR	pk + ct (kB)
RQC-Block-AG-128	47	85	2	3	28	4	4	4	4	4	170	176	-133	1.53
RQC-Block-AG-192	61	123	2	4	43	4	5	5	5	5	246	232	-198	2.86

Fig. 13: Parameters for RQC-Block-AG, $\omega = 2.8$, $1 \in \text{Supp } \mathbf{R}_2$

Likewise for the ILRPC scheme, one also proposes parameters which achieve 128 and 192 bits of security against attacks with $\omega = 2$. The new resulting parameters can be found in Figure 15.

8.3 Comparison with other schemes

For comparison, we compare our sizes with those of other encryption schemes, see Figure 16. We can see that our scheme has very competitive performances for 128 bits of security, by getting slightly smaller sizes than the lattice-based scheme KYBER.

Scheme	128 bits	192 bits
RQC-Block-MS-AG (this paper, Figure 14)	1.43	2.89
RQC-Block-AG (this paper, Figure 14)	1.53	2.86
ILRPC-Block-MS (this paper, Figure 8)	1.67	3.28
KYBER [12]	1.56	2.26
BIKE [6]	3.11	6.20
RQC [1]	5.48	8.54
LowMS [9]	5.76	14.97
HQC [2]	6.73	13.56
Classic McEliece [5]	261.2	624.3

Fig. 16: Comparaison of different schemes, the sizes represent the sum of the key and the ciphertext, expressed in kB

9 Conclusion

We showed in this paper that combine the blockwise errors and multiple syndromes approach allowed to reach small parameters than the previous versions of the RQC and LRPC schemes. We also propose to decrease the decoding failure

Scheme	m	n_2	q	k	ε	r_x	r_y	r_1	r_2	r_e	n_1	DFR	pk + ct (kB)
RQC-Block-MS-AG-128 (this paper)	43	52	2	3	32	4	4	4	4	4	2	-145	1.43
RQC-Block-AG-128 (this paper)	47	85	2	3	28	4	4	4	4	4	1	-133	1.43
RQC-Block-128 [31]	83	79	2	7	-	4	4	4	4	4	1	-	2.56
RQC-NH-MS-AG-128 [18]	61	50	2	3	51	7	7	7	5	12	3	-158	2.7
RQC-128 [1]	127	113	2	3	-	7	7	7	7	13	1	-	5.48
RQC-Block-MS-AG-192 (this paper)	67	68	2	3	45	5	5	5	5	6	2	-196	2.89
RQC-Block-AG-192 (this paper)	61	119	2	3	42	4	5	5	5	5	1	-196	2.77
RQC-Block-192	127	113	2	3	-	5	5	5	5	5	1	-	5.48
RQC-NH-MS-AG-192	79	95	2	5	65	8	8	8	5	13	2	-238	4.7
RQC-192	151	149	2	5	-	8	8	8	8	16	1	-	8.54

Fig. 14: Comparaision of parameters of different RQC schemes, $\omega = 2.8$

Scheme	m	n_2	q	k	ε	r_x	r_y	r_1	r_2	r_e	n_1	Att. 1	Att. 2	Att. 3	DFR	pk + ct (kB)
RQC-Block-MS-AG-128	59	61	2	3	42	5	4	5	4	9	2	147	151	170	-138	2.28
RQC-Block-MS-AG-192	67	81	2	3	56	5	5	5	5	10	2	213	208	250	-195	3.42

Fig. 15: Parameters for RQC-Block-MS-AG schemes, $\omega = 2$

rate of the Augmented Gabidulin code $\mathcal{G}$ that we use in our RQC-Block-MS-AG scheme (see Figure 4): the error to decode being $\mathbf{x} \cdot \mathbf{R}_2 - \mathbf{y} \cdot \mathbf{R}_1 + \mathbf{E}$, one can impose 1 to be in the support of $\mathbf{R}_2$ without diminishing the practical security of the scheme. By doing this, we make sure that the support of $\mathbf{x}$ is included in the support of the error to decode. Consequently, we can deduce a subspace of the support of the error of dimension r_x , hence to reduce the parameters of the scheme: we reach 1.31kB for the sum of the public key and ciphertext.

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